Chapter 4

Linear Programming: The Simplex Method

"Checking the results of a decision against its expectations shows executives what their strengths are, where they need to improve, and where they lack knowledge or information."

– Peter Drucker

PREVIEW

The purpose of this chapter is to help you gain an understanding of how the simplex method works. Understanding the underlying principles help to interpret and analyze solution of any LP problem.

LEARNING OBJECTIVES

After studying this chapter, you should be able to

- understand the meaning of the word 'simplex' and logic of using simplex method.
- convert an LP problem into its standard form by adding slack, surplus and/or artificial variables.
- set-up simplex tables and solve LP problems using the simplex algorithm.
- interpret the optimal solution of LP problems.
- recognize special cases such as degeneracy, multiple optimal solution, unbounded and infeasible solutions.

CHAPTER OUTLINE

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4.1 INTRODUCTION

Most real-life problems when formulated as an LP model have more than two variables and therefore need a more efficient method to suggest an optimal solution for such problems. In this chapter, we shall discuss a procedure called the *simplex method* for solving an LP model of such problems. This method was developed by G B Dantzig in 1947.

The *simplex*, an important term in mathematics, represents an object in an *n*-dimensional space, connecting $n + 1$ points. In one dimension, the simplex represents a line segment connecting two points; in two dimensions, it represents a triangle formed by joining three points; in three dimensions, it represents a four-sided pyramid.

The concept of simplex method is similar to the graphical method. In the graphical method, extreme points of the feasible solution space are examined in order to search for the optimal solution that lies at one of these points. For LP problems with several variables, we may not be able to graph the feasible region, but the optimal solution will still lie at an extreme point of the many-sided, multidimensional figure (called an *n*-dimensional polyhedron) that represents the feasible solution space. The simplex method examines these extreme points in a systematic manner, repeating the same set of steps of the algorithm until an optimal solution is found. It is for this reason that it is also called the *iterative method*.

Since the number of extreme points of the feasible solution space are finite, the method assures an improvement in the value of objective function as we move from one iteration (extreme point) to another and achieve the optimal solution in a finite number of steps. The method also indicates when an unbounded solution is reached.

4.2 STANDARD FORM OF AN LP PROBLEM

Simplex method examines corner points of the feasible region, using matrix row operations, until an optimal solution is found.

The use of the simplex method to solve an LP problem requires that the problem be converted into its standard form. The standard form of the LP problem should have the following characteristics:

(i) All the constraints should be expressed as equations by adding slack or surplus and/or artificial variables.

(ii) The right-hand side of each constraint should be made non-negative (if not). This is done by multiplying both sides of the resulting constraint by -1.

(iii) The objective function should be of the maximization type.

The standard form of the LP problem is expressed as:

$$\text{Optimize (Max or Min) } Z = c_1 x_1 + c_2 x_2 + \ldots + c_n x_n + 0s_1 + 0s_2 + \ldots + 0s_m$$

subject to the linear constraints

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + s_1 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + s_2 &= b_2 \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ \cdot \quad\quad \cdot \quad\quad\quad\quad \cdot & \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n + s_m &= b_m \end{aligned}$$

and $$x_1, x_2, \ldots, x_n, s_1, s_2, \ldots, s_m \geq 0$$

This standard form of the LP problem can also be expressed in the compact form as follows:

$$\text{Optimize (Max or Min) } Z = \sum_{j=1}^{n} c_j x_j + \sum_{i=1}^{m} 0s_i \qquad \text{(Objective function)}$$

subject to the linear constraints

$$\sum_{j=1}^{n} a_{ij}x_j + s_i = b_i; \quad i = 1, 2, \ldots, m \qquad \text{(Constraints)}$$

and $$x_j, s_i \geq 0, \quad \text{for all } i \text{ and } j \qquad \text{(Non-negativity conditions)}$$

In matrix notations the standard form is expressed as:

Optimize (Max or Min) $Z = \mathbf{cx} + \mathbf{0s}$

subject to the linear constraints

$$\mathbf{Ax} + \mathbf{s} = \mathbf{b}, \text{ and } \mathbf{x}, \mathbf{s} \geq \mathbf{0}$$

where, $\mathbf{c} = (c_1, c_2, \ldots, c_n)$ is the row vector, $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T$, $\mathbf{b} = (b_1, b_2, \ldots, b_m)^T$ and $\mathbf{s} = (s_1, s_2, \ldots, s_m)$ are column vectors, and $\mathbf{A}$ is the $m \times n$ matrix of coefficients, a_{ij} of variables $x_1, x_2, \ldots, x_n$ in the constraints.

Remarks Any LP problem (maximization or minimization) may have

1. (a) no feasible solution, i.e. value of decision variables, x_j ($j = 1, 2, \ldots, n$) may not satisfy every constraint.
   (b) a unique optimum feasible solution.
   (c) more than one optimum feasible solution, i.e. alternative optimum feasible solutions.
   (d) a feasible solution for which the objective function is *unbounded* (value of the objective function can be made as large as possible in a maximization problem or as small as possible in a minimization problem).
2. Any minimization LP problem can be converted into an equivalent maximization problem by changing the sign of c_j's in the objective function. That is,

$$\text{Minimize } \sum_{j=1}^{n} c_j x_j = \text{Maximize } \sum_{j=1}^{n} (-c_j) x_j$$

3. Any constraint expressed by equality (=) sign may be replaced by two weak inequalities. For example, $a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$ is equivalent to following two simultaneous constraints,

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \leq b_1 \quad \text{and} \quad a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n \geq b_1$$

Slack variable represents an unused quantity of resource; it is added to less-than or equal-to type constraints in order to get an equality constraint.

4. Three types of additional variables, namely (i) slack variables (s) (ii) surplus variables (– s), and (iii) artificial variables (A) are added in the given LP problem to convert it into the standard form for the following reasons:
   (a) These variables allow us to convert inequalities into equalities, thereby converting the given LP problem into its standard form. Such form will help in getting solution of the LP problem.
   (b) These variables help decision-makers to draw economic interpretation from the final solution.
   (c) These variables are used to get an initial feasible solution represented by the columns of the identity matrix.

The summary of the additional variables to be added in the given LP problem in order to convert it into a standard form is given in Table 4.1.

Table 4.1 Summary of Additional Variables Added in an LP Problem

Types of Constraint	*Extra Variable Needed*	*Coefficient of Additional Variables in the Objective Function*		*Presence of Additional Variables in the Initial Solution*
		Max Z	*Min Z*	
• Less than or equal to (≤)	A slack variable is added	0	0	Yes
• Greater than or equal to (≥)	A surplus variable is subtracted, and	0	0	No
	an artificial variable is added	$-M$	$+M$	Yes
• Equal to (=)	Only an artificial variable is added	$-M$	$+M$	Yes

Remark *A slack variable* represents an unused resource, either in the form of time on a machine, labour hours, money, warehouse space, etc. Since these variables don't yield any profit, therefore such variables are added to the objective function with zero coefficients.

A surplus variable represents the amount by which solution values exceed a resource. These variables are also called *negative slack variables*. Surplus variables, like slack variables carry a zero coefficient in the objective function.

Definitions

Basic solution Given a system of m simultaneous linear equations with n ($> m$) variables: $\mathbf{Ax} = \mathbf{B}$, where $\mathbf{A}$ is an $m \times n$ matrix and rank ($\mathbf{A}$) = m. Let $\mathbf{B}$ be any $m \times m$ non-singular submatrix of $\mathbf{A}$ obtained by reordering m linearly independent columns of $\mathbf{A}$. Then, a solution obtained by setting $n - m$ variables not associated with the columns of $\mathbf{B}$, equal to zero, and solving the resulting system is called a *basic solution* to the given system of equations.

The m variables (may be all non zero) are called *basic variables.* The $m \times m$ non-singular sub-matrix $\mathbf{B}$ is called a *basis matrix* and the columns of $\mathbf{B}$ as *basis vectors.*

If $\mathbf{B}$ is the basis matrix, then the basic solution to the system of equations will be $\mathbf{x_B} = \mathbf{B}^{-1}\mathbf{b}$.

Basic feasible solution A basic solution to the system of simultaneous equations, $\mathbf{Ax} = \mathbf{b}$ is called *basic feasible* if $\mathbf{x_B} \geq 0$.

Degenerate solution A basic solution to the system of simultaneous equations, $\mathbf{Ax} = \mathbf{b}$ is called *degenerate* if one or more of the basic variables assume zero value.

Cost vector Let $\mathbf{x_B}$ be a basic feasible solution to the LP problem.

Maximize $Z = \mathbf{cx}$

subject to the constraints

$$\mathbf{Ax} = \mathbf{b}, \quad \text{and} \quad \mathbf{x} \geq 0.$$

Then the vector $\mathbf{c_B} = (c_{B1}, c_{B2}, \ldots, c_{Bm})$, is called *cost vector* that represents the coefficient of basic variable, $\boldsymbol{x_B}$ in the basic feasible solution.

4.3 SIMPLEX ALGORITHM (MAXIMIZATION CASE)

The steps of the simplex algorithm for obtaining an optimal solution (if it exists) to a linear programming problem are as follows:

Step 1: Formulation of the mathematical model

(i) Formulate the LP model of the given problem.

(ii) If the objective function is of minimization, then convert it into equivalent maximization, by using the following relationship

$$\text{Minimize } Z = -\text{Maximize } Z^*, \text{ where } Z^* = -Z.$$

(iii) Check whether all the b_i ($i = 1, 2, \ldots, m$) values are positive. If any one of them is negative, then multiply the corresponding constraint by -1 in order to make $b_i > 0$. In doing so, remember to change a $\leq$ type constraint to a $\geq$ type constraint, and vice versa.

(iv) Express the given LP problem in the standard form by adding additional variables in constraints (as per requirement) and assign a *z*ero-cost coefficient to these variables in the objective function.

(v) Replace each *unrestricted* variable (if any) with the difference of the two non-negative variables.

Surplus variable represents the amount of resource usage above the minimum required and is added to greater-than or equal-to constraints in order to get equality constraint.

Step 2: Set-up the initial solution Write down the coefficients of all the variables in the LP problem in a tabular form, as shown in Table 4.2, in order to get an initial basic feasible solution [$\mathbf{x_B} = \mathbf{B^{-1}}\,\mathbf{b}$].

		$c_j \rightarrow$	c_1	c_2	...	c_n	0	0	...	0
Basic Variables Coefficient (c_B)	*Basic Variables* B	*Basic Variables Value* b ($= x_B$)	*Variables* x_1	x_2	...	x_n	s_1	s_2	...	s_m
c_{B1}	s_1	$x_{B1} = b_1$	a_{11}	a_{12}	...	a_{1n}	1	0	...	0
c_{B2}	s_2	$x_{B2} = b_2$	a_{21}	a_{22}	...	a_{2n}	0	1	...	0
.	.	.	.	.		.	.	.		.
.	.	.	.	.		.	.	.		.
.	.	.	.	.		.	.	.		.
c_{Bm}	s_m	$x_{Bm} = b_m$	a_{m1}	a_{m2}	...	a_{mn}	0	0	...	1
$Z = \Sigma\, c_{Bi} x_{Bi}$		$z_j = \Sigma\, c_{Bi} x_j$	0	0	...	0	0	0	...	0
		$c_j - z_j$	$c_1 - z_1$	$c_2 - z_2$	...	$c_n - z_n$	0	0	...	0

Table 4.2 Initial Simplex Table

After having set up the initial simplex table, locate the identity matrix (contains all *z*eros except positive elements 1's on the diagonal). The identity matrix so obtained is also called a *basis matrix* [because basic feasible solution is represented by $\mathbf{B} = \mathbf{I}$].

The columns of identity matrix represent the coefficients of slack variables that have been added to the constraints. Each column of the identity matrix also represents a basic variable.

Assign values of the constants (b_i's) to the column variables in the identity matrix [because $\mathbf{x_B} = \mathbf{B}^{-1}\,\mathbf{b} = \mathbf{I}\,\mathbf{b} = \mathbf{b}$].

Degeneracy arises when there is a tie in the minimum ratio value that determines the variable to enter into the next solution.

The variables corresponding to the columns of the identity matrix are called *basic variables* and the remaining ones are called *non-basic variables.* In general, if an LP model has n variables and m ($< n$) constraints, then m variables would be basic and $n - m$ variables non-basic. In certain cases one or more than one basic variables may also have *z*ero values. It one or more basic variables have zero value, then this situation is called *degeneracy* and will be discussed later.

The first row in Table 4.2 indicates the coefficients c_j of variables in the objective function. These values represent the cost (or profit) per unit associated with a variable in the objective function and these are used to determine the variable to be entered into the basis matrix **B**.

The column '$\mathbf{c_B}$' lists the coefficients of the current basic variables in the objective function. These values are used to calculate the value of Z when one unit of any variable is brought into the solution. Column '$\mathbf{x_B}$' represents the values of the basic variables in the current basic solution.

Basis is the set of variables present in the solution, have positive non-zero value; these variables are also called **basic variables.**

Numbers, a_{ij} in the columns under each variable are also called *substitution rates* (or *exchange coefficients*) because these represent the rate at which resource i ($i = 1, 2, \ldots, m$) is consumed by each unit of an activity j ($j = 1, 2, \ldots, n$).

The values z_j represent the amount by which the value of objective function Z would be decreased (*or increased*) *if one unit of the given variable is added to the new solution. Each of the values in the $c_j - z_j$ row represents the net amount of increase* (*or decrease*) *in the objective function that would occur when one unit of the variable represented by the column head is introduced into the solution.* That is:

$$c_j - z_j \text{ (net effect)} = c_j \text{ (incoming unit profit/cost)} - z_j \text{ (outgoing total profit/cost)}$$

where z_j = Coefficient of basic variables column × Exchange coefficient column j

Step 3: Test for optimality Calculate the $c_j - z_j$ value for all non-basic variables. To obtain the value of z_j multiply each element under *'Variables'* column (columns, $\mathbf{a_j}$ of the coefficient matrix) with the corresponding elements in $\mathbf{c_B}$-column. Examine values of $c_j - z_j$. The following three cases may arise:

(i) If all $c_j - z_j \le 0$, then the basic feasible solution is optimal.

(ii) If at least one column of the coefficients matrix (i.e. $\mathbf{a}_k$) for which $c_k - z_k > 0$ and all other elements are negative (i.e. $a_{ik} < 0$), then there exists an unbounded solution to the given problem.

(iii) If at least one $c_j - z_j > 0$, and each of these columns have at least one positive element (i.e. a_{ij}) for some row, then this indicates that an improvement in the value of objective function Z is possible.

Non-basic variables are those which are not in the 'basis' and have zero-value.

Step 4: Select the variable to enter the basis If Case (iii) of Step 3 holds, then select a variable that has the largest $c_j - z_j$ value to enter into the new solution. That is,

$$c_k - z_k = \text{Max}\ \{(c_j - z_j);\quad c_j - z_j > 0\}$$

The column to be entered is called the *key* or *pivot* column. Obviously, such a variable indicates the largest per unit improvement in the current solution.

Step 5: Test for feasibility (variable to leave the basis) After identifying the variable to become the basic variable, the variable to be removed from the existing set of basic variables is determined. For this, each number in $\mathbf{x_B}$-column (i.e. b_i values) is divided by the corresponding (but positive) number in the key column and a row is selected for which this ratio is non-negative and minimum. This ratio is called the *replacement* (*exchange*) *ratio.* That is,

$$\frac{x_{Br}}{a_{rj}} = \text{Min}\left\{\frac{x_{Bi}}{a_{rj}};\ a_{rj} > 0\right\}$$

This ratio restricts the number of units of the incoming variable that can be obtained from the exchange. *It may be noted that the division by a negative or by a zero element in key column is not permitted.*

The row selected in this manner is called the *key* or *pivot row* and it represents the variable which will leave the solution.

The element that lies at the intersection of the key row and key column of the simplex table is called *key* or *pivot* element.

Infeasibility is the situation in which no solution satisfies all constraints of an LP problem.

Step 6: Finding the new solution

(i) If the key element is 1, then the row remains the same in the new simplex table.

(ii) If the key element is other than 1, then divide each element in the key row (including the elements in $\mathbf{x_B}$-column) by the key element, to find the new values for that row.

(iii) The new values of the elements in the remaining rows of the new simplex table can be obtained by performing elementary row operations on all rows so that all elements except the key element in the key column are *z*ero. In other words, for each row other than the key row, we use the formula:

$$\text{Number in new row} = \begin{pmatrix}\text{Number in}\\ \text{old row}\end{pmatrix} \pm \left[\begin{pmatrix}\text{Number above or below}\\ \text{key element}\end{pmatrix} \times \begin{pmatrix}\text{Corresponding number in}\\ \text{the new row, that is row}\\ \text{replaced in Step 6 (ii)}\end{pmatrix}\right]$$

The new entries in $\mathbf{c_B}$ and $\mathbf{x_B}$ columns are updated in the new simplex table of the current solution.

Step 7: Repeat the procedure Go to Step 3 and repeat the procedure until all entries in the $c_j - z_j$ row are either negative or *z*ero.

$c_j - z_j$ row values represent net profit or loss resulting from introducing one unit of any variable into the 'basis' or solution mix.

Remark The flow chart of the simplex algorithm for both the maximization and the minimization LP problem is shown in Fig. 4.1.

Example 4.1 Use the simplex method to solve the following LP problem.

Maximize $Z = 3x_1 + 5x_2 + 4x_3$

subject to the constraints

(i) $2x_1 + 3x_2 \le 8$, (ii) $2x_2 + 5x_3 \le 10$, (iii) $3x_1 + 2x_2 + 4x_3 \le 15$

and $x_1, x_2, x_3 \ge 0$

[*Rewa MSc* (*Maths*)*, 1995; Meerut MSc* (*Stat.*)*, 1996; MSc* (*Maths*)*, 1998; Kurukshetra, MSc* (*Stat.*)*, 1998*]

Solution Step 1: Introducing non-negative slack variables s_1, s_2 and s_3 to convert the given LP problem into its standard form:

Maximize $Z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$

subject to the constraints

(i) $2x_1 + 3x_2 + s_1 = 8$, (ii) $2x_2 + 5x_3 + s_2 = 10$, (iii) $3x_1 + 2x_2 + 4x_3 + s_3 = 15$

and $x_1, x_2, x_3, s_1, s_2, s_3 \ge 0$

Step 2: Since all b_i (RHS values) > 0, (i = 1, 2, 3), therefore choose the initial basic feasible solution as:

$$x_1 = x_2 = x_3 = 0; s_1 = 8, s_2 = 10, s_3 = 15 \text{ and Max } Z = 0$$

This solution can also be read from the initial simplex Table 4.3 by equating row-wise values in the basis (**B**) column and solution values ($\mathbf{x_B}$) column.

Step 3: To see whether the current solution given in Table 4.3 is optimal or not, calculate

$$c_j - z_j = c_j - \mathbf{c_B}\,\mathbf{B^{-1}}\,\mathbf{a}_j = c_j - \mathbf{c_B}\,\mathbf{y}_j$$

for non-basic variables x_1, x_2 and x_3 as follows.

z_j = (Basic variable coefficients, $\mathbf{c_B}$) × (jth column of data matrix)

That is,

$z_1 = 0\,(2) + 0\,(0) + 0\,(3) = 0$ for x_1-column

$z_2 = 0\,(3) + 0\,(2) + 0\,(2) = 0$ for x_2-column

$z_3 = 0\,(0) + 0\,(5) + 0\,(4) = 0$ for x_3-column

These z_j values are now subtracted from c_j values in order to calculate the net profit likely to be gained by introducing (performing) a particular variable (activity) x_1, x_2 or x_3 into the new solution mix. This is done by introducing one unit of each variable x_1, x_2 and x_3 into the new solution mix.

$$c_1 - z_1 = 3 - 0 = 3, \qquad c_2 - z_2 = 5 - 0 = 5, \qquad c_3 - z_3 = 4 - 0 = 4$$

The z_j and $c_j - z_j$ rows are added into the Table 4.3.

The values of basic variables, s_1, s_2 and s_3 are given in the *solution values* ($\mathbf{x_B}$) column of Table 4.3. The remaining variables that are non-basic at the current solution have zero value. The value of objective function at the current solution is given by

$$Z = (\text{Basic variables coefficient, } \mathbf{c_B}) \times (\text{Basic variables value, } \mathbf{x_B})$$
$$= 0\,(8) + 0\,(10) + 0\,(15) = 0$$

Table 4.3
Initial Solution

		$c_j \rightarrow$	3	5	4	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\,(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	*Min Ratio* x_B/x_2
0	s_1	8	2	(3)	0	1	0	0	8/3 →
0	s_2	10	0	2	5	0	1	0	10/2
0	s_3	15	3	2	4	0	0	1	15/2
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	3	5 ↑	4	0	0	0	

Since all $c_j - z_j \geq 0$ ($j = 1, 2, 3$), the current solution is not optimal. Variable x_2 is chosen to enter into the basis because $c_2 - z_2 = 5$ is the largest positive number in the x_2-column, where all elements are positive. This means that for every unit of variable x_2, the objective function will increase in value by 5. The x_2-column is the key column.

Step 4: The variable that is to leave the basis is determined by dividing the values in the $\mathbf{x_B}$-column by the corresponding elements in the key column as shown in Table 4.3. Since the ratio, 8/3 is minimum in row 1, the basic variable s_1 is chosen to leave the solution (basis).

Step 5 (***Iteration 1***): Since the key element enclosed in the circle in Table 4.3 is not 1, divide all elements of the key row by 3 in order to obtain new values of the elements in this row. The new values of the elements in the remaining rows for the new Table 4.4 can be obtained by performing the following elementary row operations on all rows so that all elements except the key element 1 in the key column are zero.

$$R_1 \text{ (new)} \rightarrow R_1 \text{ (old)} \div 3 \text{ (}\textit{key element}\text{)}$$
$$\rightarrow (8/3,\ 2/3,\ 3/3,\ 0/3,\ 1/3,\ 0/3,\ 0/3) = (8/3,\ 2/3,\ 1,\ 0,\ 1/3,\ 0,\ 0)$$

R_2 (new) $\rightarrow R_2$ (old) $- 2R_1$ (new)
$10 - 2 \times 8/3 = 14/3$
$0 - 2 \times 2/3 = -4/3$
$2 - 2 \times 1 = 0$
$5 - 2 \times 0 = 5$
$0 - 2 \times 1/3 = -2/3$
$1 - 2 \times 0 = 1$
$0 - 2 \times 0 = 0$

R_3 (new) $\rightarrow R_3$ (old) $- 2R_1$ (new)
$15 - 2 \times 8/3 = 29/3$
$3 - 2 \times 2/3 = 5/3$
$2 - 2 \times 1 = 0$
$4 - 2 \times 0 = 4$
$0 - 2 \times 1/3 = -2/3$
$0 - 2 \times 0 = 0$
$1 - 2 \times 0 = 1$

	$c_j \rightarrow$		3	5	4	0	0	0	
Basics Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	Min Ratio x_B/x_3
5	x_2	8/3	2/3	1	0	1/3	0	0	–
0	s_2	14/3	– 4/3	0	(5)	– 2/3	1	0	(14/3)/5 →
0	s_3	29/3	5/3	0	4	– 2/3	0	1	(29/3)/4
$Z = 40/3$		z_j	10/3	5	0	5/3	0	0	
		$c_j - z_j$	– 1/3	0	4 ↑	– 5/3	0	0	

Table 4.4 Improved Solution

The improved basic feasible solution can be read from Table 4.4 as: $x_2 = 8/3$, $s_2 = 14/3$, $s_3 = 29/3$ and $x_1 = x_3 = s_1 = 0$. The improved value of the objective function becomes:

$$Z = (\text{Basic variable coefficients, } \mathbf{c_B}) \times (\text{Basic variable values, } \mathbf{x_B})$$
$$= 5\,(8/3) + 0\,(14/3) + 0\,(29/3) = 40/3$$

Once again, calculate values of $c_j - z_j$ to check whether the solution shown in Table 4.4 is optimal or not. Since $c_3 - z_3 > 0$, the current solution is not optimal.

Iteration 2: Repeat Steps 3 to 5. Table 4.5 is obtained by performing following row operations to enter variable x_3 into the basis and to drive out s_2 from the basis.

$$R_2 \text{ (new)} = R_2 \text{ (old)} \div 5 \text{ (}\textit{key element}\text{)} = (14/15, -4/15, 0, 1, -2/15, 1/5, 0)$$

R_3 (new) $\rightarrow$ R_3 (old) $- 4R_2$ (new)
$29/3 - 4 \times 14/15 = 89/15$
$5/3 - 4 \times -4/15 = 41/15$
$0 - 4 \times 0 = 0$
$4 - 4 \times 1 = 0$
$-2/3 - 4 \times -2/15 = -2/15$
$0 - 4 \times 1/5 = -4/5$
$1 - 4 \times 0 = 0$

Table 4.5 is completed by calculating the new z_j and $c_j - z_j$ values and the new value of objective function:

$z_1 = 5\,(2/3) + 4\,(-4/15) + 0\,(41/15) = 34/15$ $\quad$ $c_1 - z_1 = 3 - 34/15 = 11/15$ for x_1-column

$z_4 = 5\,(1/3) + 4\,(-2/15) + 0\,(2/15) = 17/15$ $\quad$ $c_4 - z_4 = 0 - 17/15 = -17/15$ for s_1-column

$z_5 = 5\,(0) + 4\,(1/5) + 0\,(-4/5) = 4/5$ $\quad$ $c_5 - z_5 = 0 - 4/5 = -4/5$ for s_2-column

The new objective function value is:

$$Z = (\text{Basic variables coefficient, } \mathbf{c_B}) \times (\text{Basic variables value, } \mathbf{x_B})$$
$$= 5\,(8/3) + 4\,(14/15) + 0\,(89/15) = 256/15$$

The improved basic feasible solution is shown in Table 4.5.

	$c_j \rightarrow$		3	5	4	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	Min Ratio x_B/x_1
5	x_2	8/3	2/3	1	0	1/3	0	0	(8/3)/(2/3) = 4
4	x_3	14/15	– 4/15	0	1	– 2/15	1/5	0	–
0	s_3	89/15	(41/15)	0	0	2/15	– 4/5	1	(89/15)/(41/15) = 2.17 →
$Z = 256/15$		z_j	34/15	5	4	17/15	4/5	0	
		$c_j - z_j$	11/15 ↑	0	0	– 17/15	– 4/5	0	

Table 4.5 Improved Solution

Iteration 3: In Table 4.5, since $c_1 - z_1$ is still a positive value, the current solution is not optimal. Thus, the variable x_1 enters the basis and s_3 leaves the basis. To get another improved solution as shown in Table 4.5 perform the following row operations in the same manner as discussed earlier.

R_3 (new) $\rightarrow R_3$ (old) $\times$ 15/41 (*key element*)
$\rightarrow$ (89/15 $\times$ 15/41, 41/15 $\times$ 15/41, 0 $\times$ 15/41, 0 $\times$ 15/41,
$-$ 2/15 $\times$ 15/41, $-$ 4/5 $\times$ 15/41, 1 $\times$ 15/41)
$\rightarrow$ (89/41, 1, 0, 0, $-$ 2/41, $-$ 12/41, 15/41)

R_1 (new) $\rightarrow R_1$ (old) $-$ (2/3) R_3 (new)				R_2 (new) $\rightarrow R_2$ (old) + (4/15) R_3 (new)			
8/3	$-$ 2/3 $\times$	89/3	= 50/41	14/15	+ 4/15 $\times$	89/41	= 62/41
2/3	$-$ 2/3 $\times$	1	= 0	$-$ 4/15	+ 4/15 $\times$	1	= 0
1	$-$ 2/3 $\times$	0	= 1	0	+ 4/15 $\times$	0	= 0
0	$-$ 2/3 $\times$	0	= 0	1	+ 4/15 $\times$	0	= 1
1/3	$-$ 2/3 $\times$	$-$ 2/41	= 15/41	$-$ 2/15	+ 4/15 $\times$	$-$ 2/41	= $-$ 6/41
0	$-$ 2/3 $\times$	$-$ 2/41	= 8/41	1/5	+ 4/15 $\times$	$-$ 12/41	= 5/41
0	$-$ 2/3 $\times$	15/41	= $-$ 10/41	0	+ 4/15 $\times$	15/41	= 4/41

Table 4.6 Optimal Solution

		$c_j \rightarrow$	3	5	4	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b (= x_B)	x_1	x_2	x_3	s_1	s_2	s_3
5	x_2	50/41	0	1	0	15/41	8/41	$-$ 10/41
4	x_3	62/41	0	0	1	$-$ 6/41	5/41	4/41
3	x_1	89/41	1	0	0	$-$ 2/41	$-$ 12/41	15/41
Z = 765/41		z_j	3	5	4	45/41	24/41	11/41
		$c_j - z_j$	0	0	0	$-$ 45/41	$-$ 24/41	$-$ 11/41

In Table 4.6, all $c_j - z_j < 0$ for non-basic variables. Therefore, the optimal solution is reached with, x_1 = 89/41, x_2 = 50/41, x_3 = 62/41 and the optimal value of Z = 765/41.

Example 4.2 A company makes two kinds of leather belts, belt A and belt B. Belt A is a high quality belt and belt *B* is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. The production of each of type *A* requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B.

What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the simplex method.

Solution Let x_1 and x_2 be the number of belts of type A and B, respectively, manufactured each day. Then the LP model would be as follows:

Maximize (total profit) $Z = 4x_1 + 3x_2$

subject to the constraints

(i) $2x_1 + x_2 \le 1{,}000$ (Time availability), (ii) $x_1 + x_2 \le 800$ (Supply of leather)

(iii) $\left.\begin{matrix} x_1 \le 400 \\ x_2 \le 700 \end{matrix}\right\}$ (Buckles availability)

and $x_1, x_2 \ge 0$

Standard form Introducing slack variables s_1, s_2, s_3 and s_4 to convert given LP model into its standard form as follows.

Maximize $Z = 4x_1 + 3x_2 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

subject to the constraints

(i) $2x_1 + x_2 + s_1 = 1{,}000$, (ii) $x_1 + x_2 + s_2 = 800$

(iii) $x_1 + s_3 = 400$, (iv) $x_2 + s_4 = 700$

and $x_1, x_2, s_1, s_2, s_3, s_4 \geq 0$

Solution by simplex method An initial feasible solution is obtained by setting $x_1 = x_2 = 0$. Thus, the initial solution is: $s_1 = 1{,}000$, $s_2 = 800$, $s_3 = 400$, $s_4 = 700$ and Max $Z = 0$. This solution can also be read from the initial simplex Table 4.7.

	$c_j \rightarrow$		4	3	0	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	Min Ratio x_B/x_1
0	s_1	1,000	2	1	1	0	0	0	1,000/2 = 500
0	s_2	800	1	1	0	1	0	0	800/1 = 800
0	s_3	400	(1)	0	0	0	1	0	400/1 = 400 →
0	s_4	700	0	1	0	0	0	1	not defined
$Z = 0$		z_j	0	0	0	0	0	0	
		$c_j - z_j$	4 ↑	3	0	0	0	0	

Table 4.7 Initial Solution

In Table 4.7, since $c_1 - z_1 = 4$ is the largest positive number, we apply the following row operations in order to get an improved basic feasible solution by entering variable x_1 into the basis and removing variable s_3 from the basis.

R_3 (new) $\rightarrow R_3$ (old) $\div$ 1 (*key element*) R_1 (new) $\rightarrow R_1$ (old) $- 2R_3$ (new)

R_2 (new) $\rightarrow R_2$ (old) $- R_3$ (new)

The new solution is shown in Table 4.8.

	$c_j \rightarrow$		4	3	0	0	0	0	
Basic Variables Coefficient c_B	Basis Variables B	Basic Variables Value b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	Min Ratio x_B/x_2
0	s_1	200	0	(1)	1	0	− 2	0	200/1 = 200 →
0	s_2	400	0	1	0	1	− 1	0	400/1 = 400
4	x_1	400	1	0	0	0	1	0	–
0	s_4	700	0	1	0	0	0	1	700/1 = 700
Z = 1,600		z_j	4	0	0	0	4	0	
		$c_j - z_j$	0	3 ↑	0	0	− 4	0	

Table 4.8 An Improved Solution

The solution shown in Table 4.8 is not optimal because $c_2 - z_2 > 0$ in x_2-column. Thus, again applying the following row operations to get a new solution by entering variable x_2 into the basis and removing variable s_1 from the basis, we get

R_1 (new) $\rightarrow R_1$ (old) $\div$ 1 (*key element*) R_2 (new) $\rightarrow R_2$ (old) $- R_1$ (new)

R_4 (new) $\rightarrow R_4$ (old) $- R_1$ (new)

The improved solution is shown in Table 4.9.

Table 4.9
Improved Solution

		$c_j \rightarrow$	4	3	0	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4	*Min Ratio* x_B/s_3
3	x_2	200	0	1	1	0	−2	0	—
0	s_2	200	0	0	−1	1	(1)	0	200/1 = 200 →
4	x_1	400	1	0	0	0	1	0	400/1 = 400
0	s_4	500	0	0	−1	0	2	1	500/2 = 250
Z = 2,200		z_j	4	3	3	0	−2	0	
		$c_j - z_j$	0	0	−3	0	2	0	
							↑		

The solution shown in Table 4.9 is not optimal because $c_5 - z_5 > 0$ in s_3-column. Thus, again applying the following row operations to get a new solution by entering variable s_3, into the basis and removing variable s_2 from the basis, we get

R_2 (new) $\rightarrow R_2$ (old) ÷ 1 (*key element*) $\qquad R_1$ (new) $\rightarrow R_1$ (old) + $2R_2$ (new)

R_3 (new) $\rightarrow R_3$ (old) − R_2 (new); $\qquad R_4$ (new) $\rightarrow R_4$ (old) − $2R_2$ (new)

The new improved solution is shown in Table 4.10.

Table 4.10
Optimal Solution

		$c_j \rightarrow$	4	3	0	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* b $(= x_B)$	x_1	x_2	s_1	s_2	s_3	s_4
3	x_2	600	0	1	− 1	2	0	0
0	s_3	200	0	0	− 1	1	1	0
4	x_1	200	1	0	1	− 1	0	0
0	s_4	100	0	0	1	− 2	0	1
Z = 2,600		z_j	4	3	1	2	0	0
		$c_j - z_j$	0	0	− 1	− 2	0	0

Since all $c_j - z_j < 0$ correspond to non-basic variables columns, the current basic feasible solution is also the optimal solution. Thus, the company must manufacture, x_1 = 200 belts of type A and x_2 = 600 belts of type B in order to obtain the maximum profit of Rs 2,600.

Example 4.3 A pharmaceutical company has 100 kg of A, 180 kg of B and 120 kg of C ingredients available per month. The company can use these materials to make three basic pharmaceutical products namely 5-10-5, 5-5-10 and 20-5-10, where the numbers in each case represent the percentage of weight of A, B and C, respectively, in each of the products. The cost of these raw materials is as follows:

Ingredient	*Cost per kg (Rs)*
A	80
B	20
C	50
Inert ingredients	20

The selling prices of these products are Rs 40.5, Rs 43 and 45 per kg , respectively. There is a capacity restriction of the company for product 5-10-5, because of which the company cannot produce more than 30 kg per month. Determine how much of each of the products the company should produce in order to maximize its monthly profit. [*Delhi Univ., MBA, 2004, AMIE, 2005*]

Solution Let the P_1, P_2 and P_3 be the three products to be manufactured. The data of the problem can then be summarized as follows:

Product	*Product Ingredients*			*Inert*
	A	*B*	*C*	
P_1	5%	10%	5%	80%
P_2	5%	5%	10%	80%
P_3	20%	5%	10%	65%
Cost per kg (Rs)	80	20	50	20

Cost of $P_1 = 5\% \times 80 + 10\% \times 20 + 5\% \times 50 + 80\% \times 20 = 4 + 2 + 2.50 + 16 =$ Rs 24.50 per kg

Cost of $P_2 = 5\% \times 80 + 5\% \times 20 + 10\% \times 50 + 80\% \times 20 = 4 + 1 + 5 + 16 =$ Rs 26 per kg

Cost of $P_3 = 20\% \times 80 + 5\% \times 20 + 10\% \times 50 + 65\% \times 20 = 16 + 1 + 5 + 13 =$ Rs 35 per kg

Let x_1, x_2 and x_3 be the quantity (in kg) of P_1, P_2 and P_3, respectively to be manufactured. The LP problem can then be formulated as

Maximize (net profit) Z = (Selling price – Cost price) × (Quantity of product)

$$= (40.50 - 24.50)\,x_1 + (43 - 26)\,x_2 + (45 - 35)\,x_3 = 16x_1 + 17x_2 + 10x_3$$

subject to the constraints

$$\frac{1}{20}x_1 + \frac{1}{20}x_2 + \frac{1}{5}x_3 \le 100 \quad \text{or} \quad x_1 + x_2 + 4x_3 \le 2{,}000$$

$$\frac{1}{10}x_1 + \frac{1}{20}x_2 + \frac{1}{20}x_3 \le 180 \quad \text{or} \quad 2x_1 + x_2 + x_3 \le 3{,}600$$

$$\frac{1}{20}x_1 + \frac{1}{10}x_2 + \frac{1}{10}x_3 \le 120 \quad \text{or} \quad x_1 + 2x_2 + 2x_3 \le 2{,}400$$

$$x_1 \le 30$$

and $x_1, x_2, x_3 \ge 0$.

Standard form Introducing slack variables s_1, s_2 and s_3 to convert the given LP model into its standard form as follows:

Maximize $Z = 16x_1 + 17x_2 + 10x_3 + 0s_1 + 0s_2 + 0s_3 + 0s_4$

subject to the constraints

(i) $x_1 + x_2 + 4x_3 + s_1 = 2{,}000$, (ii) $2x_1 + x_2 + x_3 + s_2 = 3{,}600$

(iii) $x_1 + 2x_2 + 2x_3 + s_3 = 2{,}400$, (iv) $x_1 + s_4 = 30$

and $x_1, x_2, x_3, s_1, s_2, s_3, s_4 \ge 0$

Solution by simplex method An initial basic feasible solution is obtained by setting $x_1 = x_2 = x_3 = 0$. Thus, the initial solution shown in Table 4.11 is: $s_1 = 2{,}000$, $s_2 = 3{,}600$, $s_3 = 2{,}400$, $s_4 = 30$ and Max $Z = 0$.

	$c_j \rightarrow$	*16*	*17*	*10*	*0*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basis Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4	*Min Ratio* x_B/x_2
0	s_1	2,000	1	1	4	1	0	0	0	2,000/1 = 2,000
0	s_2	3,600	2	1	1	0	1	0	0	3,600/1 = 3,600
0	s_3	2,400	1	(2)	2	0	0	1	0	2,400/2 = 1,200 →
0	s_4	30	1	0	0	0	0	0	1	—
$Z = 0$		z_j	0	0	0	0	0	0	0	
		$c_j - z_j$	16	17 ↑	10	0	0	0	0	

Table 4.11
Initial Solution

Since $c_2 - z_2 = 17$ in x_2-column is the largest positive value, we apply the following row operations in order to get a new improved solution by entering variable x_2 into the basis and removing variable s_3 from the basis.

R_3 (new) → R_3 (old) ÷ 2 (*key element*) R_1 (new) → R_1 (old) – R_3 (new)

R_2 (new) → R_2 (old) – R_3 (new)

The new solution is shown in Table 4.12.

Table 4.12 Improve Solution

	$c_j \rightarrow$	16	17	10	0	0	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4	Min Ratio x_B/x_1
0	s_1	800	1/2	0	3	1	0	– 1/2	0	800/(1/2) = 1,600
0	s_2	2,400	3/2	0	0	0	1	– 1/2	0	2,400/(3/2) = 1,600
17	x_2	1,200	1/2	1	1	0	0	1/2	0	1,200/(1/2) = 2,400
0	s_4	30	(1)	0	0	0	0	0	1	30/1 = 30 →
Z = 20,400		z_j	17/2	17	17	0	0	17/2	0	
		$c_j - z_j$	15/2 ↑	0	– 7	0	0	– 17/2	0	

The solution shown in Table 4.12 is not optimal because $c_1 - z_1 > 0$ in x_1-column. Thus, applying the following row operations to get a new improved solution by entering variable x_1 into the basis and removing the variable s_4 from the basis, we get

R_4 (new) $\rightarrow R_4$ (old) ÷ 1 (*key element*); R_1 (new) $\rightarrow R_1$ (old) – (1/2) R_4 (new)

R_2 (new) $\rightarrow R_2$ (old) – (3/2) R_4 (new); R_3 (new) $\rightarrow R_3$ (old) – (1/2) R_4 (new)

The new solution is shown in Table 4.13.

Table 4.13 Optimal Solution

	$c_j \rightarrow$	16	17	10	0	0	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	s_4
0	s_1	785	0	0	3	1	0	– 1/2	– 1/2
0	s_2	2,355	0	0	0	0	1	– 1/2	– 3/2
17	x_2	1,185	0	1	1	0	0	1/2	– 1/2
16	x_1	30	1	0	0	0	0	0	1
Z = 20,625		z_j	16	17	17	0	0	17/2	15/2
		$c_j - z_j$	0	0	– 7	0	0	– 17/2	– 15/2

Since all $c_j - z_j < 0$ corresponding to non-basic variables columns, the current solution is an optimal solution. Thus, the company must manufacture, x_1 = 30 kg of P_1, x_2 = 1,185 kg of P_2 and x_3 = 0 kg of P_3 in order to obtain the maximum net profit of Rs 20,625.

4.4 SIMPLEX ALGORITHM (MINIMIZATION CASE)

In certain cases, it is difficult to obtain an initial basic feasible solution of the given LP problem. Such cases arise

(i) when the constraints are of the ≤ type,

$$\sum_{j=1}^{n} a_{ij}\, x_j \le b_i, \quad x_j \ge 0$$

and value of few right-hand side constants is negative [i.e. $b_i < 0$]. After adding the non-negative slack variable s_i (i = 1, 2, . . ., m), the initial solution so obtained will be $s_i = -b_i$ for a particular resource, i. This solution is not feasible because it does not satisfy non-negativity conditions of slack variables (i.e. $s_i \ge 0$).

(ii) when the constraints are of the ≥ type,

$$\sum_{j=1}^{n} a_{ij}\, x_j \ge b_i, \quad x_j \ge 0$$

After adding surplus (negative slack) variable s_i, the initial solution so obtained will be $-s_i = b_i$ or $s_i = -b_i$

$$\sum_{j=1}^{n} a_{ij}\, x_j - s_i = b_i, \quad x_j \geq 0,\ s_i \geq 0$$

This solution is not feasible because it does not satisfy non-negativity conditions of surplus variables (i.e. $s_i \geq 0$). In such a case, artificial variables, A_i ($i = 1, 2, \ldots, m$) are added to get an initial basic feasible solution. The resulting system of equations then becomes:

$$\sum_{j=1}^{n} a_{ij}\, x_j - s_i + A_i = b_i$$

$$x_j, s_i, A_i \geq 0, \quad i = 1, 2, \ldots, m$$

These are m simultaneous equations with $(n + m + m)$ variables (n decision variables, m artificial variables and m surplus variables). An initial basic feasible solution of LP problem with such constraints can be obtained by equating $(n + 2m - m) = (n + m)$ variables equal to zero. Thus the new solution to the given LP problem is: $A_i = b_i$ ($i = 1, 2, \ldots, m$), which is not the solution to the original system of equations because the two systems of equations are not equivalent. Thus, to get back to the original problem, artificial variables must be removed from the optimal solution. There are two methods for removing artificial variables from the solution.

- *Two-Phase Method*
- *Big-M Method or Method of Penalties*

The simplex method, both for the minimization and the maximization LP problem, may be summarized through a flow chart shown in Fig. 4.1.

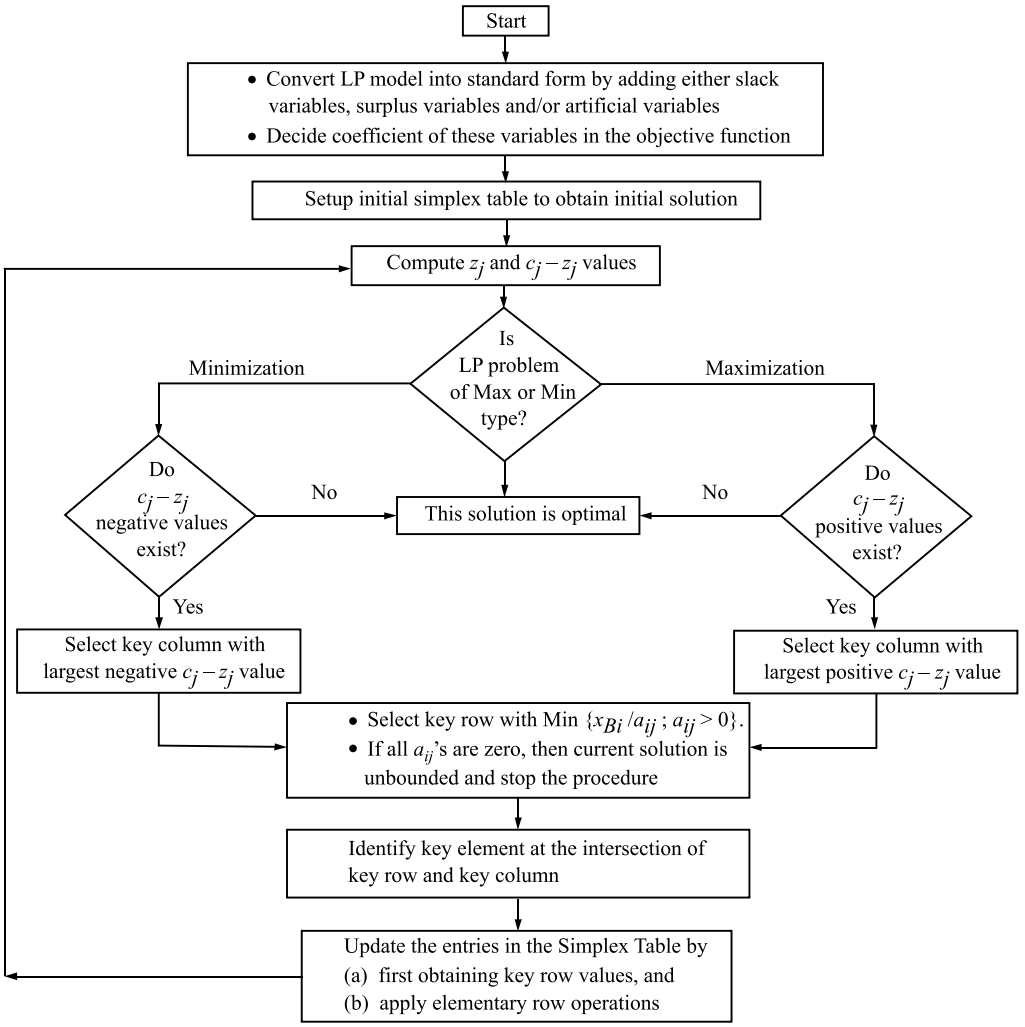


Fig. 4.1 Flow Chart of Simplex Algorithm

Remark Artificial variables have no meaning in a physical sense and are only used as a tool for generating an initial solution to an LP problem. Before the optimal solution is reached, all artificial variables must be dropped out from the solution mix. This is done by assigning appropriate coefficients to these variables in the objective function. These variables are added to those constraints with equality (=) and greater than or equal to (≥) sign.

4.4.1 Two-Phase Method

In the first phase of this method, the sum of the artificial variables is minimized subject to the given constraints in order to get a basic feasible solution of the LP problem. The second phase minimizes the original objective function starting with the basic feasible solution obtained at the end of the first phase. Since the solution of the LP problem is completed in two phases, this method is called the *two-phase method.*

Advantages of the method

1. No assumptions on the original system of constraints are made, i.e. the system may be redundant, inconsistent or not solvable in non-negative numbers.
2. It is easy to obtain an initial basic feasible solution for Phase I.
3. The basic feasible solution (if it exists) obtained at the end of phase I is used as initial solution for Phase II.

An artificial variable is added to the constraints to get an initial solution to an LP problem.

Steps of the Algorithm: *Phase I*

Step 1 (a): If all the constraints in the given LP problem are 'less than or equal to' (≤) type, then Phase II can be directly used to solve the problem. Otherwise, the necessary number of surplus and artificial variables are added to convert constraints into equality constraints.

(b) If the given LP problem is of minimization, then convert it to the maximization type by the usual method.

Step 2: Assign zero coefficient to each of the decision variables (x_j) and to the surplus variables; and assign – 1 coefficient to each of the artificial variables. This yields the following auxiliary LP problem.

$$\text{Maximize } Z^* = \sum_{i=1}^{m} (-1)\, A_i$$

subject to the constraints

$$\sum_{i=1}^{n} a_{ij}\, x_j + A_i = b_i, \qquad i = 1, 2, \ldots, m$$

and $$x_j, A_j \geq 0$$

Step 3: Apply the simplex algorithm to solve this auxiliary LP problem. The following three cases may arise at optimality.

(a) Max $Z^* = 0$ and at least one artificial variable is present in the basis with positive value. This means that no feasible solution exists for the original LP problem.

(b) Max $Z^* = 0$ and no artificial variable is present in the basis. This means that only decision variables (x_j's) are present in the basis and hence proceed to Phase II to obtain an optimal basic feasible solution on the original LP problem.

(c) Max $Z^* = 0$ and at least one artificial variable is present in the basis at *z*ero value. This means that a feasible solution to the auxiliary LP problem is also a feasible solution to the original LP problem. In order to arrive at the basic feasible solution, proceed directly to Phase II or else eliminate the artificial basic variable and then proceed to Phase II.

Remark Once an artificial variable has left the basis, it has served its purpose and can, therefore, be removed from the simplex table. An artificial variable is never considered for re-entry into the basis.

Phase II: Assign actual coefficients to the variables in the objective function and *z*ero coefficient to the artificial variables which appear at *z*ero value in the basis at the end of Phase I. The last simplex table of

Phase I can be used as the initial simplex table for Phase II. Then apply the usual simplex algorithm to the modified simplex table in order to get the optimal solution to the original problem. Artificial variables that do not appear in the basis may be removed.

Example 4.4 Use two-phase simplex method to solve the following LP problem:

Minimize $Z = x_1 + x_2$

subject to the constraints

(i) $2x_1 + x_2 \geq 4$, (ii) $x_1 + 7x_2 \geq 7$

and $x_1, x_2 \geq 0$

Solution Converting the given LP problem objective function into the maximization form and then adding surplus variables s_1 and s_2 and artificial variables A_1 and A_2 in the constraints, the problem becomes:

Maximize $Z^* = -x_1 - x_2$

subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 4$, (ii) $x_1 + 7x_2 - s_2 + A_2 = 7$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

where $Z^* = -Z$

Phase I: This phase starts by considering the following auxiliary LP problem:

Maximize $Z^* = -A_1 - A_2$

subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 4$, (ii) $x_1 + 7x_2 - s_2 + A_2 = 7$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

The initial solution is presented in Table 4.14.

	$c_j \rightarrow$	*0*	*0*	*0*	*0*	*– 1*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2
–1	A_1	4	2	1	– 1	0	1	0
– 1	A_2	7	1	(7)	0	– 1	0	1 →
$Z^* = -11$		z_j	– 3	– 8	1	1	– 1	– 1
		$c_j - z_j$	3	8 ↑	– 1	– 1	0	0

Table 4.14 Initial Solution

Artificial variables A_1 and A_2 are now removed, one after the other, maintaining the feasibility of the solution.

Iteration 1: Applying the following row operations to get an improved solution by entering variable x_2 in the basis and first removing variable A_2 from the basis. The improved solution is shown in Table 4.15. Note that the variable x_1 cannot be entered into the basis as this would lead to an infeasible solution.

R_2 (new) $\rightarrow R_2$ (old) $\div$ 7 (*key element*); R_1 (new) $\rightarrow R_1$ (old) $- R_2$ (new)

	$c_j \rightarrow$	*0*	*0*	*0*	*0*	*– 1*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2*
– 1	A_1	3	13/7	0	– 1	(1/7)	1	– 1/7 →
0	x_2	1	1/7	1	0	– 1/7	0	1/7
$Z^* = -3$		z_j	– 13/7	0	1	– 1/7	– 1	1/7
		$c_j - z_j$	13/7	0	– 1	1/7 ↑	0	– 8/7

Table 4.15 Improved Solution

* This column can permanently be removed at this stage.

Iteration 2: To remove A_1 from the solution shown in Table 4.15, we enter variable s_2 in the basis by applying the following row operations. The new solution is shown in Table 4.16. It may be noted that if instead of s_2, variable x_1 is chosen to be entered into the basis, it will lead to an infeasible solution

$$R_1 \text{ (new)} \to R_1 \text{ (old)} \times 7; \qquad R_2 \text{ (new)} \to R_2 \text{ (old)} + (1/7)\, R_1 \text{ (new)}$$

Table 4.16 Improved Solution

		$c_j \to$	0	0	0	0	−1	−1
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2	A_1^*	A_2^*
0	s_2	21	13	0	−7	1	7	−1
0	x_2	4	2	1	−1	0	1	0
$Z^* = 0$		z_j	0	0	0	0	0	0
		$c_j - z_j$	0	0	0	0	−1	−1

* Remove columns A_1 and A_2 from Table 4.16.

Since all $c_j - z_j \le 0$ correspond to non-basic variables, the optimal solution: $x_1 = 0$, $x_2 = 4$, $s_1 = 0$, $s_2 = 21$, $A_1 = 0$, $A_2 = 0$ with $Z^* = 0$ is arrived at. However, this solution may or may not be the basic feasible solution to the original LP problem. Thus, we have to move to Phase II to get an optimal solution to our original LP problem.

Phase II: The modified simplex table obtained from Table 4.16 is represented in Table 4.17.

Table 4.17

		$c_j \to$	−1	−1	0	0	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2	Min Ratio x_B/x_1
0	s_2	21	(13)	0	−7	1	21/13 →
−1	x_2	4	2	1	−1	0	4/2
$Z^* = -4$		z_j	−2	−1	1	0	
		$c_j - z_j$	1 ↑	0	−1	0	

Iteration 1: Introducing variable x_1 into the basis and removing variable s_2 from the basis by applying the following row operations:

$$R_1 \text{ (new)} \to R_1 \text{ (old)} \div 13 \text{ (\textit{key element})}; \qquad R_2 \text{ (new)} \to R_2 \text{ (old)} - 2R_1 \text{ (new)}$$

The improved basic feasible solution so obtained is given in Table 4.18. Since in Table 4.18, $c_j - z_j \le 0$ for all non-basic variables, the current solution is optimal. Thus, the optimal basic feasible solution to the given LP problem is: $x_1 = 21/13$, $x_2 = 10/13$ and Max $Z^* = -31/13$ or Min $Z = 31/13$.

Table 4.18 Optimal Solution

		$c_j \to$	−1	−1	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(=x_B)$	x_1	x_2	s_1	s_2
−1	x_1	21/13	1	0	−7/13	1/13
−1	x_2	10/13	0	1	1/13	−2/13
$Z^* = -31/13$		z_j	−1	−1	6/13	1/13
		$c_j - z_j$	0	0	−6/13	−1/13

Example 4.5 Solve the following LP problem by using the two-phase simplex method.

Minimize $Z = x_1 - 2x_2 - 3x_3$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 = 2$, (ii) $2x_1 + 3x_2 + 4x_3 = 1$

and $x_1, x_2, x_3 \ge 0$. [*Meerut, MSc (Math), 2003*]

Solution After converting the objective function into the maximization form and by adding artificial variables A_1 and A_2 in the constraints, the given LP problem becomes:

Maximize $Z^* = -x_1 + 2x_2 + 3x_3$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 + A_1 = 2$, (ii) $2x_1 + 3x_2 + 4x_3 + A_2 = 1$

and $x_1, x_2, x_3, A_1, A_2 \geq 0$

where $Z^* = -Z$

Phase I: This phase starts by considering the following auxiliary LP problem:

Maximize $Z^* = -A_1 - A_2$

subject to the constraints

(i) $-2x_1 + x_2 + 3x_3 + A_1 = 2$ (ii) $2x_1 + 3x_2 + 4x_3 + A_2 = 1$

and $x_1, x_2, x_3, A_1, A_2 \geq 0$

The initial solution is presented in Table 4.19.

		$c_j \rightarrow$	0	0	0	– 1	– 1
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\,(= x_B)$	x_1	x_2	x_3	A_1	A_2
– 1	A_1	2	– 2	1	3	1	0
– 1	A_2	1	2	3	(4)	0	1 →
$Z^* = -3$		z_j	0	– 4	– 7	– 1	– 1
		$c_j - z_j$	0	4	7 ↑	0	0

Table 4.19 Initial Solution

To first remove the artificial variable A_2 from the solution shown in Table 4.19, introduce variable x_3 into the basis by applying the following row operations:

$$R_1 \text{ (new)} \rightarrow R_2 \text{ (old)} \div 4 \text{ (}key\ element\text{)}; \quad R_1 \text{ (new)} \rightarrow R_1 \text{ (old)} - 3R_2 \text{ (new)}$$

The improved solution so obtained is given in Table 4.20. Since in Table 4.20, $c_j - z_j \leq 0$ corresponds to non-basic variables, the optimal solution is: $x_1 = 0$, $x_2 = 0$, $x_3 = 1/4$, $A_1 = 5/4$ and $A_2 = 0$ with Max $Z^* = -5/4$. But at the same time, the value of $Z^* < 0$ and the artificial variable A_1 appears in the basis with positive value 5/4. Hence, feasible solution to the given original LP problem does not exist.

		$c_j \rightarrow$	0	0	0	–1
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	A_1
– 1	A_1	5/4	–7/2	–5/4	0	1
0	x_3	1/4	1/2	3/4	1	0
$Z^* = -5/4$		z_j	7/2	5/4	0	– 1
		$c_j - z_j$	– 7/2	– 5/4	0	0

Table 4.20 Optimal but not Feasible Solution

Example 4.6 Use two-phase simplex method to solve following LP problem.

Maximize $Z = 3x_1 + 2x_2 + 2x_3$

subject to the constraints

(i) $5x_1 + 7x_2 + 4x_3 \leq 7$, (ii) $-4x_1 + 7x_2 + 5x_3 \geq -2$, (iii) $3x_1 + 4x_2 - 6x_3 \geq 29/7$

and $x_1, x_2, x_3 \geq 0$. [*Punjab Univ., BE (E & C) 2006 ; BE (IT) 2006*]

Solution Since RHS of constraint 2 is negative, multiplying it by – 1 on both sides and express it as: $4x_1 - 7x_2 - 5x_3 \leq 2$.

Phase I: Introducing slack, surplus and artificial variables in the constraints, the standard form of LP problem becomes:

Minimize $Z^* = A_1$

subject to the constraints

(i) $5x_1 + 7x_2 + 4x_3 + s_1 = 7$, (ii) $4x_1 - 7x_2 - 5x_3 + s_2 = 2$, (iii) $3x_1 + 4x_2 - 6x_3 - s_3 + A_1 = 29/7$

and $x_1, x_2, x_3, s_1, s_2, s_3, A_1 \geq 0$.

Setting variables $x_1 = x_2 = x_3 = s_3 = 0$, the basic feasible solution shown in Table 4.21 to the auxiliary LP problem is: $s_1 = 7, s_2 = 2, A_1 = 29/7$

Table 4.21 Initial Solution

		$c_j \rightarrow$	0	0	0	0	0	0	1	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_2
0	s_1	7	5	(7)	4	1	0	0	0	1 →
0	s_2	2	4	−7	−5	0	1	0	0	—
1	A_1	29/7	3	4	−6	0	0	−1	1	29/28
$Z = 29/7$		z_j	3	4	−6	0	0	−1	1	
		$c_j - z_j$	−3	−4 ↑	6	0	0	1	0	

Since $c_2 - z_2 = -4$ is the largest negative value under x_2-column in Table 4.21, replacing basic variable s_1 with the non-basic variable x_2 into the basis. For this, apply following row operations:

$R_1(\text{new}) \rightarrow R_1(\text{old}) \div 7(\textit{key element})$; $R_2(\text{new}) \rightarrow R_2(\text{old}) + 7R_1(\text{new})$;

$R_3(\text{new}) \rightarrow R_3(\text{old}) - 4R_1(\text{new})$

to get an improved solution as shown in Table 4.22.

Table 4.22 An Improved Solution

		$c_j \rightarrow$	0	0	0	0	0	0	1	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_1
0	x_2	1	5/7	1	4/7	1/7	0	0	0	7/5
0	s_2	9	9	0	−1	1	1	0	0	1 } Tie
1	A_1	1/7	(1/7)	0	−58/7	−4/7	0	−1	1	1 } Tie
$Z = 1/7$		z_j	1/7	0	−58/7	−4/7	0	−1	1	
		$c_j - z_j$	−1/7 ↑	0	58/7	4/7	0	1	0	

Solution shown in Table 4.22 is not optimal and there is a tie between the s_2 and A_1-rows for the key row. The A_1-row is selected as the key row because preference is given to the artificial variable. Thus (1/7) is the key element. Replace basic variable A_1 in the basis with the non-basic variable x_1. Applying following row operations to get the new solution as shown in Table 4.23.

$R_3(\text{new}) \rightarrow R_3(\text{old}) \div (1/7)$ key element; $R_1(\text{new}) \rightarrow R_1(\text{old}) - (5/7)\, R_3\,(\text{new})$;

$R_2(\text{new}) \rightarrow R_2(\text{old}) - 9\, R_3(\text{new})$

Table 4.23 Basic Feasible Solution

		$c_j \rightarrow$	0	0	0	0	0	0
Basic Variables Coefficient c_B	*Variables in Basis* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
0	x_2	2/7	0	1	42	3	0	5
0	s_2	0	0	0	521	37	1	63
0	x_1	1	1	0	−58	−4	0	−7
Z = 0		z_j	0	0	0	0	0	0
		$c_j - z_j$	0	0	0	0	0	0

In Table 4.23, all $c_j - z_j = 0$ under non-basic variable columns. Thus, the current solution is optimal. Also Min $Z^* = 0$ and no artificial variable appears in the basis, and hence this solution is the basic feasible solution to the original LP problem.

Phase II: Using the solution shown in Table 4.23 as the initial solution for Phase II and carrying out computations to get optimal solution as shown in Table 4.24.

		$c_j \rightarrow$	3	2	2	0	0	0	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3	*Min Ratio* x_B/x_3
2	x_2	2/7	0	1	42	3	0	5	1/147
0	s_2	0	0	0	521	37	1	63	0 →
3	x_1	1	1	0	– 58	– 6	0	– 7	—
$Z = 25/7$		z_j	3	2	– 90	– 6	0	– 11	
		$c_j - z_j$	0	0	92 ↑	6	0	11	

Table 4.24 Initial Solution

In Table 4.24, $c_3 - z_3 = 92$ is the largest positive value corresponding non-basic variable x_3, replacing basic variable s_2 with non-basic variable x_3 into the basis. For this apply necessary row operations as usual. The new solution is shown in Table 4.25.

		$c_j \rightarrow$	3	2	2	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	s_3
2	x_2	2/7	0	1	0	9/521	– 42/521	– 41/521
2	x_3	0	0	0	1	37/521	1/521	63/521
3	x_1	1	1	0	0	62/521	58/521	7/521
Z = 25/7		z_j	3	2	2	278/521	92/521	65/521
		$c_j - z_j$	0	0	0	–278/521	– 92/521	– 65/521

Table 4.25 Optimal Solution

Since all $c_j - z_j \leq 0$ in Table 4.25, the current solution is the optimal basic feasible solution: $x_1 = 1, x_2 = 2/7$, $x_3 = 0$ and Max Z = 25/7.

4.4.2 Big-M Method

Big-M method is another method of removing artificial variables from the basis. In this method, large undesirable (unacceptable penalty) coefficients to artificial variables are assigned from the point of view of the objective function. If the objective function Z is to be minimized, then a very large positive price (called *penalty*) is assigned to each artificial variable. Similarly, if Z is to be maximized, then a very large negative price (also called *penalty*) is assigned to each of these variables. The penalty is supposed to be designated by $– M$, for a maximization problem, and $+ M$, for a minimization problem, where $M > 0$. The Big-M method for solving an LP problem can be summarized in the following steps:

Step 1: Express the LP problem in the standard form by adding slack variables, surplus variables and/or artificial variables. Assign a *z*ero coefficient to both slack and surplus variables. Then assign a very large coefficient $+ M$ (minimization case) and $– M$ (maximization case) to artificial variable in the objective function.

Step 2: The initial basic feasible solution is obtained by assigning zero value to decision variables, x_1, x_2, ..., etc.

Step 3: Calculate the values of $c_j - z_j$ in last row of the simplex table and examine these values.

(i) If all $c_j - z_j \geq 0$, then the current basic feasible solution is optimal.

(ii) If for a column, k, $c_k - z_k$ is most negative and all entries in this column are negative, then the problem has an unbounded optimal solution.

(iii) If one or more $c_j - z_j < 0$ (minimization case), then select the variable to enter into the basis (solution mix) with the largest negative $c_j - z_j$ value (largest per unit reduction in the objective function value). This value also represents the opportunity cost of not having one unit of the variable in the solution. That is,

$$c_k - z_k = \text{Min } \{c_j - z_j : c_j - z_j < 0\}$$

Step 4: Determine the key row and key element in the same manner as discussed in the simplex algorithm for the maximization case.

Step 5: Continue with the procedure to update solution at each iteration till optimal solution is obtained.

Remarks At any iteration of the simplex algorithm one of the following cases may arise:

1. If at least one artificial variable is a basic variable (i.e., variable that is present in the basis) with zero value and the coefficient it M in each $c_j - z_j$ ($j = 1, 2, \ldots, n$) values is non-negative, then the given LP problem has no solution. That is, the current basic feasible solution is degenerate.
2. If at least one artificial variable is present in the basis with a positive value and the coefficients M in each $c_j - z_j$ ($j = 1, 2, \ldots, n$) values is non-negative, then the given LP problem has no optimum basic feasible solution. In this case, the given LP problem has a *pseudo optimum* basic feasible solution.

Example 4.7 Use penalty (Big-M) method to solve the following LP problem.

Minimize $Z = 5x_1 + 3x_2$

subject to the constraints

$$\text{(i) } 2x_1 + 4x_2 \le 12, \quad \text{(ii) } 2x_1 + 2x_2 = 10, \quad \text{(iii) } 5x_1 + 2x_2 \ge 10$$

and $x_1, x_2 \ge 0$.

Solution Adding slack variable, s_1; surplus variable, s_2 and artificial variables, A_1 and A_2 in the constraints of the given LP problem, the standard form of the LP problem becomes.

Minimize $Z = 5x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$

subject to the constraints

$$\text{(i) } 2x_1 + 4x_2 + s_1 = 12, \quad \text{(ii) } 2x_1 + 2x_2 + A_1 = 10 \quad \text{(iii) } 5x_1 + 2x_2 - s_2 + A_2 = 10$$

and $x_1, x_2, s_1, s_2, A_1, A_2 \ge 0$

An initial basic feasible solution: $s_1 = 12$, $A_1 = 10$, $A_2 = 10$ and Min $Z = 10M + 10M = 20M$ is obtained by putting $x_1 = x_2 = s_2 = 0$. It may be noted that the columns that correspond to the current basic variables and form the basis (identity matrix) are s_1 (slack variable), A_1 and A_2 (both artificial variables). The initial basic feasible solution is given in Table 4.26.

Since the value $c_1 - z_1 = 5 - 7M$ is the smallest value, therefore variable x_1 is chosen to enter into the basis (solution mix). To decide a current basic variable to leave the basis, calculate minimum ratio as shown in Table 4.26.

		$c_j \rightarrow$	5	3	0	0	M	M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b (= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2	*Min Ratio* x_B/x_1
0	s_1	12	2	4	1	0	0	0	12/2 = 6
M	A_1	10	2	2	0	0	1	0	10/2 = 5
M	A_2	10	(5)	2	0	– 1	0	1	10/5 = 2 →
$Z = 20M$		z_j	$7M$	$4M$	0	$-M$	M	M	
		$c_j - z_j$	$5 - 7M$ ↑	$3 - 4M$	0	M	0	0	

Table 4.26 Initial Solution

Iteration 1: Introduce variable x_1 into the basis and remove A_2 from the basis by applying the following row operations.

R_3 (new) $\rightarrow R_3$ (old) ÷ 5 (*key element*); R_2 (new) $\rightarrow R_2$ (old) $- 2R_3$ (new).

R_1 (new) $\rightarrow R_1$ (old) $- 2R_3$ (new).

The improved basic feasible solution is shown in Table 4.27.

		$c_j \rightarrow$	5	3	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/x_2
0	s_1	8	0	(16/5)	1	2/5	0	8/(16/5) = 5/2 →
M	A_1	6	0	6/5	0	2/5	1	6/(6/5) = 5
5	x_1	2	1	2/5	0	– 1/5	0	2/(2/5) = 5
$Z = 10 + 6M$		z_j	5	$(6M/5) + 2$	0	$(2M/5) - 1$	M	
		$c_j - z_j$	0	$(-6M/5) + 1$ ↑	0	$(-2M/5) + 1$	0	

Table 4.27 Improved Solution

Iteration 2: Since the value of $c_2 - z_2$ in Table 4.27 is the largest negative value, variable x_2 is chosen to replace basic variable s_1 in the basis. Thus, to get an improved basic feasible solution, apply, the following row operations:

R_1 (new) $\rightarrow R_1$ (old) × 5/16 (*key element*); R_2 (new) $\rightarrow R_2$ (old) – (6/5) R_1 (new).

R_3 (new) $\rightarrow R_3$ (old) – (2/5) R_1 (new).

The new solution is shown in Table 4.28.

		$c_j \rightarrow$	5	3	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/s_2
3	x_2	5/2	0	1	5/16	1/8	0	(5/2)/(1/8) = 40
M	A_1	3	0	0	– 3/8	(1/4)	1	3/(1/4) = 12 →
5	x_1	1	1	0	– 1/8	– 1/4	0	
$Z = 25/2 + 3M$		z_j	5	$3 - 3M/8 + 5/16$		$M/4 - 7/8$	M	
		$c_j - z_j$	0	$03M/8 - 5/16$		$-M/4 + 7/8$ ↑	0	

Table 4.28 Improved Solution

Iteration 3: Since $c_4 - z_4 < 0$ (negative) in s_2-column, the current solution is not optimal. Thus, non-basic variable s_2 is chosen to replace artificial variable A_1 in the basis. To get an improved basic feasible solution, apply the following row operations:

R_2 (new) $\rightarrow R_2$ (old) × 4 (*key element*); R_1 (new) $\rightarrow R_1$ (old) – (1/8) R_2 (new)

R_3 (new) $\rightarrow R_3$ (old) + (1/4) R_2 (new).

The improved basic feasible solution is shown in Table 4.29.

		$c_j \rightarrow$	5	3	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2
3	x_2	1	0	1	1/2	0
0	s_2	12	0	0	– 3/2	1
5	x_1	4	1	0	– 1/2	0
$Z = 23$		z_j	5	3	– 1	0
		$c_j - z_j$	0	0	1	0

Table 4.29 Optimal Solution

In Table 4.24, all $c_j - z_j \geq 0$. Thus an optimal solution is arrived at with the value of variables as: $x_1 = 4$, $x_2 = 1$, $s_1 = 0$, $s_2 = 12$ and Min $Z = 23$.

Example 4.8 Use penalty (Big-M) method to solve the following LP problem.

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

subject to the constraints

(i) $x_1 + 2x_2 + 3x_3 = 15$, (ii) $2x_1 + x_2 + 5x_3 = 20$, (iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4 \geq 0$ [*Calicut, BTech. (Engg), 2000; Bangalore BE (Mech.), 2000; AMIE, 2009*]

Solution Since all constraints of the given LP problem are equations, therefore only artificial variables A_1 and A_2 are added in the constraints to convert given LP problem to its standard form. The standard form of the problem is stated as follows:

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4 - MA_1 - MA_2$

subject to the constraints

(i) $x_1 + 2x_2 + 3x_3 + A_1 = 15$, (ii) $2x_1 + x_2 + 5x_3 + A_2 = 20$

(iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4, A_1, A_2 \geq 0$

An initial basic feasible solution is given in Table 4.30.

Table 4.30 Initial Solution

		$c_j \rightarrow$	1	2	3	–1	– M	– M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	x_3	x_4	A_1	A_2	Min Ratio x_B/x_3
– M	A_1	15	1	2	3	0	1	0	15/3 = 5
– M	A_2	20	2	1	(5)	0	0	1	20/5 = 4 →
– 1	x_4	10	1	2	1	1	0	0	10/1 = 10
$Z = -35M - 10$		z_j	$-3M-1$	$-3M-2$	$-8M-1$	–1	$-M$	$-M$	
		$c_j - z_j$	$3M+2$	$3M+4$	$8M+4$ ↑	0	0	0	

Since value of $c_3 - z_3$ in Table 4.30 is the largest positive value, the non-basic variable x_3 is chosen to replace the artificial variable A_2 in the basis. Thus, to get an improved solution, apply the following row operations.

R_2 (new) $\rightarrow R_1$ (old) ÷ 5 (*key element*) ; R_1 (new) $\rightarrow R_1$ (old) – 3 R_2 (new).

R_3 (new) $\rightarrow R_3$ (old) – R_2 (new)

The improved basic feasible solution is shown in Table 4.31.

Table 4.31 Improved Solution

		$c_j \rightarrow$	1	2	3	– 1	– M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	x_3	x_4	A_1	Min Ratio x_B/x_2
– M	A_1	3	– 1/5	(7/5)	0	0	1	$\frac{3}{7/5} = \frac{15}{7}$ →
3	x_3	4	2/5	1/5	1	0	0	$\frac{4}{1/5} = 20$
– 1	x_4	6	3/5	9/5	0	1	0	$\frac{6}{9/5} = \frac{30}{9}$
$Z = -3M + 6$		z_j	$M/5 + 3/5$	$-7M/5 - 6/5$	3	– 1	$-M$	
		$c_j - z_j$	$-M/5 - 2/5$	$7M/5 + 16/5$ ↑	0	0	0	

Since value of $c_2 - z_2 > 0$ (positive) in Table 4.31, the non-basic variable x_2 is chosen to replace the artificial variable A_1 in the basis. Thus, to get an improved basic feasible solution, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) $\times$ (5/7) (*key element*); R_2 (new) $\rightarrow R_2$ (old) – (1/5) R_1 (new).

R_3 (new) $\rightarrow R_3$ (old) – (9/5) R_1 (new).

The improved basic feasible solution is shown in Table 4.32.

		$c_j \rightarrow$	*1*	*2*	*3*	*– 1*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	x_4	*Min Ratio* x_B/x_1
2	x_2	15/7	– 1/7	1	0	0	–
3	x_3	25/7	3/7	0	1	0	25/7 × 7/3 = 25/3
– 1	x_4	15/7	(6/7)	0	0	1	15/7 × 7/6 = 15/6 →
Z = 90/7		z_j	1/7	2	3	–1	
		$c_j - z_j$	6/7 ↑	0	0	0	

Table 4.32 Improved Solution

Once again, the solution shown in Table 4.32 is not optimal as $c_1 - z_1 > 0$ in x_1-column. Thus, applying the following row operations for replacing non-basic variable x_1 in the basis with basic variable x_4:

R_3 (new) $\rightarrow R_3$ (old) $\times$ (7/6) (*key element*) ; R_1 (new) $\rightarrow R_1$ (old) + (1/7) R_3 (new).

R_2 (new) $\rightarrow R_2$ (old) – (3/7) R_3 (new).

The improved basic feasible solution is shown in Table 4.33.

		$c_j \rightarrow$	*1*	*2*	*3*	*– 1*
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	x_3	x_4
2	x_2	15/6	0	1	0	1/6
3	x_3	15/6	0	0	1	– 3/6
1	x_1	15/6	1	0	0	7/6
Z = 15		z_j	1	2	3	0
		$c_j - z_j$	0	0	0	– 1

Table 4.33 Optimal Solution

In Table 4.28, since all $c_j - z_j \leq 0$, therefore, an optimal solution is arrived at with value of variables as: x_1 = 15/6, x_2 = 15/6, x_3 = 15/6 and Max Z = 15.

Example 4.9 ABC Printing Company is facing a tight financial squeeze and is attempting to cut costs wherever possible. At present it has only one printing contract, and luckily, the book is selling well in both the hardcover and the paperback editions. It has just received a request to print more copies of this book in either the hardcover or the paperback form. The printing cost for the hardcover books is Rs 600 per 100 books while that for paperback is only Rs 500 per 100. Although the company is attempting to economize, it does not wish to lay off any employee. Therefore, it feels obliged to run its two printing presses – I and II, at least 80 and 60 hours per week, respectively. Press I can produce 100 hardcover books in 2 hours or 100 paperback books in 1 hour. Press II can produce 100 hardcover books in 1 hour or 100 paperbacks books in 2 hours. Determine how many books of each type should be printed in order to minimize costs.

Solution Let x_1 and x_2 be the number of batches containing 100 hard cover and paperback books, to be printed respectively. The LP problem can then be formulated as follows.

Minimize $Z = 600x_1 + 500x_2$

subject to the constraints

(i) $2x_1 + x_2 \geq 80$, (ii) $x_1 + 2x_2 \geq 60$ (Printing press hours)

and $x_1, x_2 \geq 0$

Standard form Adding surplus variables s_1, s_2 and artificial variables A_1, A_2 in the the constraints, the standard form of the LP problem becomes

Minimize $Z = 600x_1 + 500x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$
subject to the constraints

(i) $2x_1 + x_2 - s_1 + A_1 = 80$, (ii) $x_1 + 2x_2 - s_2 + A_2 = 60$

and $x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$

Solution by simplex method The initial basic feasible solution: $A_1 = 80, A_2 = 60$ and Min $Z = 80M + 60M = 140M$ obtained by putting $x_1 = x_2 = s_1 = s_2 = 0$ is shown in Table 4.34.

Table 4.34 Initial Solution

		$c_j \rightarrow$	600	500	0	0	M	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	A_2	Min Ratio x_B/x_2
M	A_1	80	2	1	− 1	0	1	0	80/1
M	A_2	60	1	(2)	0	− 1	0	1	60/2 →
$Z = 140M$		z_j	$3M$	$3M$	$-M$	$-M$	M	M	
		$c_j - z_j$	$600 - 3M$	$500 - 3M$ ↑	M	M	0	0	

Since $c_2 - z_2$ value in x_2-column of Table 4.34 is the largest negative, therefore non-basic variable x_2 is chosen to replace basic variable A_2 in the basis. For this, apply following row operations.

R_2 (new) = R_2 (old) ÷ 2 (*key element*); R_1 (new) → R_1 (old) − R_2 (new)

to get an improved basic feasible solution as shown in Table 4.35.

Table 4.35 Improved Solution

		$c_j \rightarrow$	600	500	0	0	M	
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2	A_1	Min Ratio x_B/x_1
M	A_1	50	(3/2)	0	− 1	1/2	1	100/3 →
500	x_2	30	1/2	1	0	− 1/2	0	60
$Z = 15{,}000 + 50M$		z_j	$3M/2 + 250$	500	$-M$	$M/2 - 250$	M	
		$c_j - z_j$	$350 - 3M/2$ ↑	0	M	$250 - M/2$	0	

Again, since value $c_1 - z_1$ of in x_1-column of Table 4.35 is the largest negative, non-basic variable x_1 is chosen to replace basic variable A_1 in the basis. For this, apply following row operations:

R_1 (new) → R_1 (old) × (2/3) (*key element*) ; R_2 (new) → R_2 (old) − (1/2) R_1 (new)

to get an improved basic feasible solution as shown in Table 4.36.

Table 4.36 Optimal Solution

		$c_j \rightarrow$	600	500	0	0
Basic Variables Coefficient c_B	Basic Variables B	Basic Variables Value $b(= x_B)$	x_1	x_2	s_1	s_2
600	x_1	100/3	1	0	− 2/3	1/3
500	x_2	40/3	0	1	1/3	− 2/3
$Z = 80{,}000/3$		z_j	600	500	− 700/3	− 400/3
		$c_j - z_j$	0	0	700/3	400/3

In Table 4.36, all $c_j - z_j \geq 0$ and no artificial variable is present in the basis (solution mix). Hence, an optimum solution is arrived at with x_1 = 100/3 batches of hardcover books, x_2 = 40/3 batches of paperback books, at a total minimum cost, Z = Rs. 80,000/3.

Example 4.10 An advertising agency wishes to reach two types of audiences: Customers with annual income greater than Rs 15,000 (target audience A) and customers with annual income less than Rs 15,000 (target audience B). The total advertising budget is Rs 2,00,000. One programme of TV advertising costs Rs 50,000; one programme on radio advertising costs Rs 20,000. For contract reasons, at least three programmes ought to be on TV, and the number of radio programmes must be limited to five. Surveys indicate that a single TV programme reaches 4,50,000 customers in target audience A and 50,000 in target

audience B. One radio programme reaches 20,000 in target audience A and 80,000 in target audience *B*. Determine the media mix to maximize the total reach. [*Delhi Univ*., *MBA*, *2000*]

Solution Let x_1 and x_2 be the number of insertions in TV and radio, respectively. The LP problem can then be formulated as follows:

Maximize (total reach) $Z = (4{,}50{,}000 + 50{,}000)\,x_1 + (20{,}000 + 80{,}000)\,x_2$

$= 5{,}00{,}000\,x_1 + 1{,}00{,}000\,x_2 = 5x_1 + x_2$

subject to the constraints

(i) $50{,}000\,x_1 + 20{,}000\,x_2 \le 2{,}00{,}000$ or $5x_1 + 2x_2 \le 20$ (Advt. budget)

(ii) $x_1 \ge 3$ (Advt. on TV) (iii) $x_2 \le 5$ (Advt. on Radio)

and $x_1, x_2 \ge 0$.

Standard form Adding slack/surplus and/or artificial variables in the constraints of LP problem. Then standard form of given LP problem becomes

Maximize $Z = 5x_1 + x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $5x_1 + 2x_2 + s_1 = 20$, (ii) $x_1 - s_2 + A_1 = 3$, (iii) $x_2 + s_3 = 5$

and $x_1, x_2, s_1, s_2, s_3, A_1 \ge 0$

Solution by simplex method An initial basic feasible solution: $s_1 = 20$, $A_1 = 3$, $s_3 = 5$ and Max $Z = -3M$ obtained by putting $x_1 = x_2 = s_2 = 0$ is shown in simplex Table 4.37.

	$c_j \rightarrow$		*5*	*1*	*0*	*0*	*0*	*– M*	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_1
0	s_1	20	5	2	1	0	0	0	20/5 = 4
$-M$	A_2	3	(1)	0	0	– 1	0	1	3/1 = 3 →
0	s_3	5	0	1	0	0	1	0	—
$Z = -3M$		z_j	$-M$	0	0	M	0	$-M$	
		$c_j - z_j$	$M + 5$ ↑	1	0	$-M$	0	0	

Table 4.37 Initial Solution

The $c_1 - z_1$ value in x_1-column of Table 4.37 is the largest positive value. The non-basic variable x_1 is chosen to replace basic variable A_1 in the basis. For this apply following row operations

R_2 (new) $\rightarrow R_2$ (old) ÷ 1 (*key element*); R_1 (new) $\rightarrow R_1$ (old) $- 5R_2$ (new)

to get the improved basic feasible solution as shown in Table 4.38.

	$c_j \rightarrow$		*5*	*1*	*0*	*0*	*0*	
Basic Variables Coefficient c_B	*Basic Variable* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3	*Min Ratio* x_B/s_2
0	s_1	5	0	2	1	(5)	0	5/5 = 1 →
5	x_1	3	1	0	0	– 1	0	—
0	s_3	5	0	1	0	0	1	—
$Z = 15$		z_j	5	0	0	– 5	0	
		$c_j - z_j$	0	1	0	5 ↑	0	

Table 4.38 Improved Solution

Again in Table 4.38, $c_4 - z_4$ value in s_2-column is the largest positive. Thus non-basic variable s_2 is chosen to replace basic variable s_1 into the basis. For this apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) ÷ 5 (*key element*); R_2 (new) $\rightarrow R_2$ (old) + R_1 (new)

to get the improved basic feasible solution as shown in Table 4.39.

Table 4.39
Optimal Solution

		$c_j \rightarrow$	5	1	0	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b(= x_B)$	x_1	x_2	s_1	s_2	s_3
0	s_2	1	0	2/5	1/5	1	0
5	x_1	4	1	2/5	1/5	0	0
0	s_3	5	0	1	0	0	1
$Z = 20$		z_j	5	2	1	0	0
		$c_j - z_j$	0	− 1	− 1	0	0

Since all $c_j - z_j \geq 0$ in Table 4.39, the total reach of target audience cannot be increased further. Hence, the optimal solution is: $x_1 = 4$ insertions in TV and $x_2 = 0$ in radio with Max (total audience) $Z = 20{,}00{,}000$.

Example 4.11 An Air Force is experimenting with three types of bombs P, Q and R in which three kinds of explosives, viz., A, B and C will be used. Taking the various factors into account, it has been decided to use the maximum 600 kg of explosive A, at least 480 kg of explosive B and exactly 540 kg of explosive C. Bomb P requires 3, 2, 2 kg, bomb Q requires 1, 4, 3 kg and bomb R requires 4, 2, 3 kg of explosives A, B and C respectively. Bomb P is estimated to give the equivalent of a 2 ton explosion, bomb Q, a 3 ton explosion and bomb R, a 4 ton explosion respectively. Under what production schedule can the Air Force make the biggest bang?

Solution Let x_1, x_2 and x_3 be the number of bombs of type P, Q and R to be experimented, respectively.

Then the LP problem can be formulated as:

Maximize $Z = 2x_1 + 3x_2 + 4x_3$

subject to the constraints

(i) Explosive A : $3x_1 + x_2 + 4x_3 \leq 600$, (ii) Explosive B : $2x_1 + 4x_2 + 2x_3 \geq 480$,

(iii) Explosive C : $2x_1 + 3x_2 + 3x_3 = 540$,

and $x_1, x_2, x_3 \geq 0$.

Standard form Adding slack, surplus and artificial variables in the constraints of LP problem, the standard form of LP problem becomes:

Maximize $Z = 2x_1 + 3x_2 + 4x_3, + 0s_1 + 0s_2 - MA_1 - MA_2$

subject to the constraints

(i) $3x_1 + x_2 + 4x_3 + s_1 = 600$, (ii) $2x_1 + 4x_2 + 2x_3 - s_2 + A_1 = 480$, (iii) $2x_1 + 3x_2 + 3x_3 + A_2 = 540$,

and $x_1, x_2, x_3 s_1, s_2, A_1, A_2 \geq 0$.

Solution by simplex method The initial basic feasible solution: $s_1 = 600$, $A_1 = 480$, $A_2 = 540$, Max $Z = -1{,}020$ M obtained by putting basic variables $x_1 = x_2 = s_2 = 0$ is shown in Table 4.40.

Table 4.40
Initial Solution

		$c_j \rightarrow$	2	3	4	0	0	–M	–M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (= x_B)$	x_1	x_2	x_3	s_1	s_2	A_1	A_2	*Min Ratio* x_B/x_2
0	s_1	600	3	1	4	1	0	0	0	600/1
$-M$	A_1	480	2	(4)	2	0	–1	1	0	480/4 →
$-M$	A_2	540	2	3	3	0	0	0	1	540/3
$Z = 1{,}020\ M$		z_j	$-4M$	$-7M$	$-5M$	0	M	$-M$	$-M$	
		$c_j - z_j$	$2 + 4M$	$3 + 7M$ ↑	$4 + 5M$	0	$-M$	0	0	

Since in Table 4.40, $c_2 - z_2$ value in x_2–column is largest positive, non-basic variable x_2 is chosen to replace basic variable A_1 in the basis. For this, apply the following row operations:

$R_2(\text{new}) \rightarrow R_2(\text{old}) \times (1/4)$ (*key element*); $R_1(\text{new}) \rightarrow R_1(\text{old}) - R_2(\text{new})$;

$R_3(\text{new}) \rightarrow R_3(\text{old}) - 3R_2(\text{new})$

to get the improved basic feasible solution as shown in Table 4.41.

	$c_j \rightarrow$	2	3	4	0	0	–M	
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	s_2	A_1	*Min Ratio* x_B/x_3
0	s_1	480	5/2	0	7/2	1	1/4	0	960/7
3	x_2	120	1/2	1	1/2	0	–1/4	0	240
–M	A_1	180	1/3	0	(3/2)	0	3/4	1	120 →
$Z = 360 - 180M$		z_j	3/2 – M/2	3	3/2 – 3M/2	0	–3/4 – 3M/4	–M	
		$c_j - z_j$	1/2 + M/2	0	5/2 + 3M/2 ↑	0	3/4 + 3M/4	0	

Table 4.41 Improved Solution

In Table 4.41, $c_3 - z_3$ value in x_3-column is largest positive, non-basic variable x_3 is chosen to replace basic variable A_2 into the basis. For this, apply following row operations:

$R_3(\text{new}) \rightarrow R_3(\text{old}) \times (2/3)$ (*key element*); $R_1(\text{new}) \rightarrow R_1(\text{old}) - (7/2)\, R_3(\text{new})$;

$R_2(\text{new}) \rightarrow R_2(\text{old}) - (1/2)\, R_3(\text{new})$.

to get the improved basic feasible solution as shown in Table 4.42.

	$c_j \rightarrow$	2	3	4	0	0
Basic Variables Coefficient c_B	*Basic Variables* B	*Basic Variables Value* $b\ (=x_B)$	x_1	x_2	x_3	s_1	s_2
0	s_1	60	4/3	0	0	1	–3/2
3	x_2	60	1/3	1	0	0	–1/2
4	x_3	120	1/3	0	1	0	1/2
$Z = 660$		z_j	7/3	3	4	0	1/2
		$c_j - z_j$	–1/3	0	0	0	–1/2

Table 4.42 Optimal Solution

In Table 4.42, all $c_j - z_j \le 0$ and artificial variables A_1 and A_2 have been removed from the basis (solution mix). Thus, an optimal solution is arrived at with $x_1 = 0$ bombs of type P, $x_2 = 60$ bombs of type Q, $x_3 = 120$ bombs of type R, at largest benefit of $Z = 660$.

SELF PRACTICE PROBLEMS A

1. A television company has three major departments for manufacturing two of its models – A and B. The monthly capacities of the departments are given as follows:

	Per Unit Time Requirement (hours)		*Hours Available this Month*
	Model A	*Model B*	
Department *I*	4.0	2.0	1,600
Department *II*	2.5	1.0	1,200
Department *III*	4.5	1.5	1,600

The marginal profit per unit from model A is Rs 400 and from model B is Rs 100. Assuming that the company can sell any quantity of either product due to favourable market conditions, determine the optimum output for both the models, the highest possible profit for this month and the slack time in the three departments.

2. A manufacturer of leather belts makes three types of belts A, B and C which are processed on three machines M_1, M_2 and M_3. Belt A requires 2 hours on machine M_1 and 3 hours on machine M_2 and 2 hours on machine M_3. Belt B requires 3 hours on machine M_1, 2 hours on machine M_2 and 2 hours on machine M_3 and Belt C requires 5 hours on machine M_2 and 4 hours on machine M_3. There are 8 hours of time per day available on machine M_1, 10 hours of time per day available on machine M_2 and 15 hours of time per day available on machine M_3. The profit gained from belt A is Rs 3.00 per unit, from Belt B is Rs 5.00 per unit, from belt C is Rs 4.00 per unit. What should be the daily production of each type of belt so that the products yield the maximum profit?

3. A company produces three products A, B and C. These products require three ores O_1, O_2 and O_3. The maximum quantities of the ores O_1, O_2 and O_3 available are 22 tonnes, 14 tonnes and 14 tonnes, respectively. For one tonne of each of these products, the ore requirements are:

	A	*B*	*C*
O_1	3	–	3
O_2	1	2	3
O_3	3	2	3
Profit per tonne (Rs in thousand)	1	4	5

The company makes a profit of Rs 1,000, 4,000 and 5,000 on each tonne of the products A, B and C, respectively. How many

tonnes of each product should the company produce in order to maximize its profits.

4. A manufacturing firm has discontinued the production of a certain unprofitable product line. This has created considerable excess production capacity. Management is considering to devote this excess capacity to one or more of three products; call them product 1, 2 and 3. The available capacity on the machines that might limit output is summarized in the following table:

Machine Type	*Available Time (in Machine-hours per Week)*
Milling Machine	250
Lathe	150
Grinder	50

The number of machine-hours required for each unit of the respective product is as follows:

Machine Type	*Productivity (in Machine-hours per Unit)*		
	Product 1	*Product 2*	*Product 3*
Milling Machine	8	2	3
Lathe	4	3	0
Grinder	2	–	1

The profit per unit would be Rs 20, Rs 6 and Rs 8, respectively for product 1, 2 and 3. Find how much of each product the firm should produce in order to maximize its profit.

[*Delhi Univ., MBA 2006*]

5. A farmer has 1,000 acres of land on which he can grow corn, wheat or soyabean. Each acre of corn costs Rs 100 for preparation, requires 7 men-days of work and yields a profit of Rs 30. An acre of wheat costs Rs 120 to prepare, requires 10 men-days of work and yields a profit of Rs 40. An acre of soyabean costs Rs 70 to prepare, requires 8 men-days of work and yields a profit of Rs 20. If the farmer has Rs 1,00,000 for preparation and can count on 8,000 men-days of work, determine how many acres should be allocated to each crop in order to maximize profits? [*Delhi Univ., MBA, 2004*]

6. The annual handmade furniture show and sale is supposed to take place next month and the school of vocational studies is also planning to make furniture for this sale. There are three wood-working classes – I year, II year and III year, at the school and they have decided to make styles of chairs – *A*, *B* and *C*. Each chair must receive work in each class. The time in hours required for each chair in each class is:

Chair	*I Year*	*II Year*	*III Year*
A	2	4	3
B	3	3	2
C	2	1	4

During the next month there will be 120 hours available to the I year class, 160 hours to the II year class, and 100 hours to the III year class for producing the chairs. The teacher of the wood-working classes feels that a maximum of 40 chairs can be sold at the show. The teacher has determined that the profit from each type of chair will be: A, Rs 40; B, Rs 35 and C, Rs. 30. How many chairs of each type should be made in order to maximize profits at the show and sale?

7. Mr Jain, the marketing manager of ABC Typewriter Company is trying to decide how he should allocate his salesmen to the company's three primary markets. Market I is in the urban area and the salesman can sell, on the average, 40 typewriters a week. Salesmen in the other two markets, II and III can sell, on the average, 36 and 25 typewriters per week, respectively. For the coming week, three of the salesmen will be on vacation, leaving only 12 men available for duty. Also because of lack of company care, a maximum of 5 salesmen can be allocated to market area I. The selling expenses per week for salesmen in each area are Rs 800 per week for area I, Rs 700 per week for area II, and Rs 500 per week for area III. The budget for the next week is Rs 7,500. The profit margin per typewriter is Rs 150. Determine how many salesmen should be assigned to each area in order to maximize profits?

8. Three products – A, B and C – are produced in three machine centres X, Y and Z. All three products require a part of their manufacturing operation at each of the machine centres. The time required for each operation on various products is indicated in the following table. Only 100, 77 and 80 hours are available at machine centres X, Y and *Z*, respectively. The profit per unit from A, B and C is Rs 12, Rs 3 and Re 1, respectively.

Products	*Machine Centres*			*Profit per Unit (Rs)*
	X	*Y*	*Z*	
A	10	7	2	12
B	2	3	4	3
C	1	2	1	1
Available hours	100	77	80	

(a) Determine suitable product mix so as to maximize the profit. Comment on the queries (b) and (c) from the solution table obtained.

(b) Satisfy that full available hours of X and Y have been utilized and there is surplus hours of Z. Find out the surplus hours of Z.

(c) Your aim is to utilize surplus capacity of Z. Can you say from the table that the introduction of more units of Y is required?

9. A certain manufacturer of screw fastenings found that there is a market for packages of mixed screw sizes. His market research data indicated that two mixtures of three screw types (1, 2 and 3), properly priced, could be most acceptable to the public. The relevant data is:

Mixture	*Specifications*	*Selling Price (Rs/kg)*
A	≥ 50% type 1 ≤ 30% type 2 and quantity of type 3	5
B	≥ 35% type 1 ≤ 45% type 2 and quantity of type 3	4

For these screws, the plant capacity and manufacturing cost are as follows:

Screw Type	*Plant Capacity (kg/day × 100)*	*Manufacturing Cost (Rs/kg)*
1	10	4.50
2	10	3.50
3	6	2.70

What production shall this manufacturer schedule for greatest profit, assuming that he can sell all that he manufactures?

10. A blender of whisky imports three grades A, B and C. He mixes them according to the recipes that specify the maximum or minimum percentages of grades A and C in each blend. These are shown in the table below.

Blend	*Specification*	*Price per Unit* (*Rs*)
Blue Dot	Not less than 60% of A Not more than 20% of C	6.80
Highland Fling	Not more than 60% of C Not less than 15% of A	5.70
Old Frenzy	Not more than 50% of C	4.50

Following are the supplies of the three whiskies along with their cost.

Whisky	*Maximum Quantity Available per Day*	*Cost per Unit* (*Rs*)
A	2,000	7.00
B	2,500	5.00
C	1,200	4.00

Show how to obtain the first matrix in a simplex computation of a production policy that will maximize profits.

11. An animal feed company must produce on a daily basis 200 kg of a mixture that consists ingredients x_1 and x_2 ingredient. x_1 costs Rs 3 per kg and x_2 costs Rs 8 per kg. Not more than 80 kg of x_1 can be used and at least 60 kg of x_2 must be used. Find out how much of each ingredient should be used if the company wants to minimize costs.

12. A diet is to contain at least 20 ounces of protein and 15 ounces of carbohydrate. There are three foods A, B and C available in the market, costing Rs 2, Re 1 and Rs 3 per unit, respectively. Each unit of A contains 2 ounces of protein and 4 ounces of carbohydrate; each unit of B contains 3 ounces of protein and 2 ounces of carbohydrate; and each unit of C contains 4 ounces of protein and 2 ounces of carbohydrate. How many units of each food should the diet contain so that the cost per unit diet is minimum?

13. A person requires 10, 12 and 12 units of chemicals A, B and C, respectively for his garden. A typical liquid product contains 5, 2 and 1 unit of A, B and C, respectively per jar. On the other hand a typical dry product contains 1, 2 and 4 units of A, B and C per unit. If the liquid product sells for Rs 3 per jar and the dry product for Rs 2 per carton, how many of each should be purchased in order to minimize the cost and meet the requirement?

14. A scrap metal dealer has received an order from a customer for a minimum of 2,000 kg of scrap metal. The customer requires that at least 1,000 kg of the shipment of metal be of high quality copper that can be melted down and used to produce copper tubing. Furthermore, the customer will not accept delivery of the order if it contains more than 175 kg metal that he deems unfit for commercial use, i.e. metal that contains an excessive amount of impurities and cannot be melted down and refined profitably.

    The dealer can purchase scrap metal from two different suppliers in unlimited quantities with following percentages (by weight) of high quality copper and unfit scrap.

	Supplier A	*Supplier B*
Copper	25%	75%
Unfit scrap	5%	10%

The cost per kg of metal purchased from supplier A and B are Re 1 and Rs 4, respectively. Determine the optimal quantities of metal to be purchased for the dealer from each of the two suppliers.

15. A marketing manager wishes to allocate his annual advertising budget of Rs 2,00,000 to two media vehicles – A and B. The unit cost of a message in media A is Rs 1,000 and that of B is Rs 1,500. Media A is a monthly magazine and not more than one insertion is desired in one issue, whereas at least five messages should appear in media B. The expected audience for unit messages in media A is 40,000 and that of media B is 55,000. Develop an LP model and solve it for maximizing the total effective audience.

16. A transistor radio company manufactures four models A, B, C and D. Models A, B and C, have profit contributions of Rs 8, Rs 15 and Rs 25 respectively and has model D a loss of Re 1. Each type of radio requires a certain amount of time for the manufacturing of components, for assembling and for packing. A dozen units of model A require one hour for manufacturing, two hours for assembling and one hour for packing. The corresponding figures for a dozen units of model B are 2, 1 and 2, and for a dozen units of C are 3, 5 and 1. A dozen units of model D however, only require 1 hour of packing. During the forthcoming week, the company will be able to make available 15 hours of manufacturing, 20 hours of assembling and 10 hours of packing time. Determine the optimal production schedule for the company.

17. A transport company is considering the purchase of new vehicles for providing transportation between the Delhi Airport and hotels in the city. There are three vehicles under consideration: Station wagons, minibuses and large buses. The purchase price would be Rs 1,45,000 for each station wagon, Rs 2,50,000 for each minibus and Rs 4,00,000 for each large bus. The board of directors has authorized a maximum amount of Rs 50,00,000 for these purchases. Because of the heavy air travel, the new vehicles would be utilized at maximum capacity, regardless of the type of vehicles purchased. The expected net annual profit would be Rs 15,000 for the station wagon, Rs 35,000 for the minibus and Rs 45,000 for the large bus. The company has hired 30 new drivers for the new vehicles. They are qualified drivers for all three types of vehicles. The maintenance department has the capacity to handle an additional 80 station wagons. A minibus is equivalent to 1.67 station wagons and each large bus is equivalent to 2 station wagons in terms of their use of the maintenance department. Determine the number of each type of vehicle that should be purchased in order to maximize profit.

18. Omega Data Processing Company performs three types of activities: Payroll, accounts receivables, and inventories. The profit and time requirement for keypunch, computation and office printing, for a standard job, are shown in the following table:

Job	*Profit/ Standard Job* (*Rs*)	*Time Requirement* (*Min*)		
		Keypunch	*Computation*	*Printing*
Payroll	275	1,200	20	100
A/c Receivable	125	1,400	15	60
Inventory	225	800	35	80

Omega guarantees overnight completion of the job. Any job scheduled during the day can be completed during the day or night. Any job scheduled during the night, however, must be completed during the night. The capacities for both day and night are shown in the following table:

Capacity (*Min*)	*Keypunch*	*Computation*	*Print*
Day	4,200	150	400
Night	9,200	250	650

Determine the mixture of standard jobs that should be accepted during the day and night.

19. A furniture company can produce four types of chairs. Each chair is first made in the carpentry shop and then furnished, waxed and polished in the finishing shop. The man-hours required in each are:

	Chair Type			
	1	*2*	*3*	*4*
Carpentry shop	4	9	7	10
Finishing shop	1	1	3	40
Contribution per chair (Rs)	12	20	18	40

The total number of man-hours available per month in carpentry and finishing shops are 6,000 and 4,000, respectively.

Assuming an abundant supply of raw material and an abundant demand for finished products, determine the number of each type of chairs that should be produced for profit maximization.

20. A metal products company produces waste cans, filing cabinets, file boxes for correspondence, and lunch boxes. Its inputs are sheet metal of two different thickness, called A and B, and manual labour. The input-output relationship for the company are shown in the table given below:

	Waste Cans	*Filing Cabinets*	*Correspondence Boxes*	*Lunch Boxes*
Sheet metal A	6	0	2	3
Sheet metal B	0	10	0	0
Manual labour	4	8	2	3

The sales revenue per unit of waste cans, filing cabinets, correspondence boxes and lunch boxes are Rs 20, Rs 400, Rs 90 and Rs 20, respectively. There are 225 units of sheet metal A available in the company's inventory, 300 of sheet metal B, and a total of 190 units of manual labour. What is the company's optimal sales revenue?

[*Delhi Univ., MBA, 2003, 2005*]

21. A company mined diamonds in three locations in the country. The three mines differed in terms of their capacities, number, weight of stones mined, and costs. These are shown in the table below:

Due to marketing considerations, a monthly production of exactly 1,48,000 stones was required. A similar requirement called for at least 1,30,000 carats (The average stone size was at least 130/148 = 0.88 carats). The capacity of each mine is measured in cubic meter. The mining costs are not included from the treatment costs and assume to be same at each mine. The problem for the company was to meet the marketing requirements at the least cost.

Mine	*Capacity (M^3 of earth processed)*	*Treatment Costs (Rs. per M^3)*	*Crude (Carats per M^3)*	*Stone Count (Number of stone per M^3)*
Plant 1	83,000	0.60	0.360	0.58
Plant 2	3,10,000	0.36	0.220	0.26
Plant 3	1,90,000	0.50	0.263	0.21

Formulate a linear programming model to determine how much should be mined at each location.

HINTS AND ANSWERS

1. Let x_1 and x_2 = units of models A and B to be manufactured, respectively.

$$\text{Max } Z = 400x_1 + 400x_2$$
$$\text{subject to } 4x_1 + 2x_2 \le 1{,}600$$
$$5x_1/2 + x_2 \le 1{,}200$$
$$9x_1/2 + 3x_2/2 \le 1{,}600$$
$$\text{and } x_1, x_2 \ge 0$$

Ans. $x_1 = 355.5$, $x_2 = 0$ and Max Z = 1,42,222.2.

2. Let x_1, x_2 and x_3 = units of types A, B and C belt to be manufactured, respectively.

$$\text{Max } Z = 3x_1 + 5x_2 + 4x_3$$
$$\text{subject to } 2x_1 + 3x_2 \le 8$$
$$2x_2 + 5x_3 \le 10$$
$$3x_1 + 2x_2 + 4x_3 \le 15$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 89/41$, $x_2 = 50/41$ $x_3 = 64/41$ and Max Z = 775/41.

3. Let x_1, x_2 and x_3 = quantity of products A, B and C to be produced, respectively.

$$\text{Max } Z = x_1 + 4x_2 + 5x_3$$
$$\text{subject to } 3x_1 + 3x_2 \le 22$$
$$x_1 + 2x_2 + 3x_3 \le 14$$
$$3x_1 + 2x_2 \le 14$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 0$, $x_2 = 7$, $x_3 = 0$ and Max Z = Rs 28,000.

4. Let x_1, x_2 and x_3 = number of units of products 1, 2 and 3 to be produced per week, respectively.

$$\text{Max } Z = 20x_1 + 6x_2 + 8x_3$$
$$\text{subject to } 8x_1 + 2x_2 + 3x_3 \le 250$$
$$4x_1 + 3x_2 \le 150$$
$$2x_1 + x_3 \le 50$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 0$, $x_2 = 50$, $x_3 = 50$ and Max Z = 700.

5. Let x_1, x_2 and x_3 = acreage of corn, wheat and soyabean, respectively.

$$\text{Max } Z = 30x_1 + 40x_2 + 20x_3$$
$$\text{subject to } 10x_1 + 12x_2 + 7x_3 \le 10{,}000$$
$$7x_1 + 10x_2 + 8x_3 \le 8{,}000$$
$$x_1 + x_2 + x_3 \le 1{,}000$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 250$, $x_2 = 625$, $x_3 = 0$ and Max Z = Rs 32,500.

6. Let x_1, x_2 and x_3 = number of units of chair of styles A, B and C, respectively.

$$\text{Max } Z = 40x_1 + 35x_2 + 30x_3$$
$$\text{subject to } 2x_1 + 3x_2 + 2x_3 \le 120$$
$$4x_1 + 3x_2 + x_3 \le 160$$
$$3x_1 + 2x_2 + 4x_3 \le 100$$
$$x_1 + x_2 + x_3 \le 40$$
$$\text{and } x_1, x_2, x_3 \ge 0$$

Ans. $x_1 = 20$, $x_2 = 20$, $x_3 = 0$ and Max Z = Rs 1,500.

7. Let x_1, x_2 and x_3 = salesman assigned to area, 1, 2 and 3, respectively.

Max. $Z = 40 \times 150x_1 + 36 \times 150x_2 + 25 \times 150x_3 - (800x_1 + 700x_2 + 500x_3)$

subject to (i) $x_1 + x_2 + x_3 \le 12$; (ii) $x_1 \le 5$;
(iii) $800x_1 + 700x_2 + 500x_3 \le 7{,}500$

and $x_1, x_2, x_3 \ge 0$

11. Let x and y = number of kg of ingredients x_1, x_2, respectively.

Min (total cost) $Z = 3x + 8y$

subject to (i) $x + y = 200$; (ii) $x \le 80$; (iii) $y \ge 60$

and $x, y \ge 0$

Ans. $x = 80$, $y = 120$ and Min Z = Rs 1,200.

12. Let x_1, x_2 and x_3 = number of units of food A, B and C, respectively which a diet must contain.

Min (total cost) $Z = 2x_1 + x_2 + x_3$

subject to $2x_1 + 2x_2 + 4x_3 \ge 20$
$4x_1 + 2x_2 + 2x_3 \ge 15$

and $x_1, x_2, x_3 \ge 0$

13. Let x_1 and x_2 = number of units of liquid and dry product produced, respectively.

Min (total cost) $Z = 3x_1 + 2x_2$

subject to (i) $5x_1 + x_2 \ge 10$; (ii) $2x_1 + 2x_2 \ge 12$;
(iii) $x_1 + 4x_2 \ge 12$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 1$, $x_2 = 5$ and Min $Z = 13$.

14. Let x_1 and x_2 = number of scrap (in kg) purchased from suppliers A and B, respectively.

Min (total cost) $Z = x_1 + 4x_2$

subject to $2.25x_1 + 0.75x_2 \ge 1{,}000$
$0.05x_1 + 0.10x_2 \ge 175$
$x_1 + x_2 \ge 2{,}000$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 2{,}500$, $x_2 = 500$ and Min $Z = 4{,}500$.

15. Let x_1 and x_2 = number of insertions messages for media A and B, respectively.

Min (total effective audience) $Z = 40{,}000x_1 + 55{,}000x_2$

subject to (i) $1{,}000x_1 + 1500x_2 \le 2{,}00{,}000$;
(ii) $x_1 \le 12$; (iii) $x_2 \ge 5$

and $x_1, x_2 \ge 0$

Ans. $x_1 = 12$, $x_2 = 16/3$ and Max $Z = 77{,}3333.33$.

16. Let x_1, x_2, x_3 and x_4 = unit of models A, B, C and D to be produce, respectively.

Max (total income) $Z = 8x_1 + 15x_2 + 25x_3 - x_4$

subject to (i) $x_1 + 2x_2 + 3x_3 = 15$; (ii) $2x_1 + x_2 + 5x_3 = 20$
(iii) $x_1 + 2x_2 + x_3 + x_4 = 10$

and $x_1, x_2, x_3, x_4 \ge 0$

Ans. $x_1 = 5/2$, $x_2 = 5/2$, $x_3 = 5/2$, $x_4 = 0$ and Max $Z = 120$.

17. Let x_1, x_2 and x_3 = number of station wagons, minibuses and large buses to be purchased, respectively.

Max $Z = 15{,}000x_1 + 35{,}000x_2 + 45{,}000x_3$

subject to $x_1 + x_2 + x_3 \le 30$
$1{,}45{,}000x_1 + 2{,}50{,}000x_2 + 4{,}00{,}000x_3 \le 50{,}00{,}000$
$x_1 + 1.67x_2 + 0.5x_3 \le 80$

and $x_1, x_2, x_3 \ge 0$

18. Let x_{ij} represents ith job and jth activity

Max $Z = 275\,(x_{11} + x_{12}) + 125(x_{21} + x_{22}) + 225\,(x_{31} + x_{32})$

subject to

$1{,}200\,(x_{11} + x_{12}) + 1{,}400\,(x_{21} + x_{22}) + 800\,(x_{31} + x_{32}) \le 13{,}400$
$20\,(x_{11} + x_{12}) + 15\,(x_{21} + x_{22}) + 35\,(x_{31} + x_{32}) \le 400$
$100\,(x_{11} + x_{12}) + 60\,(x_{21} + x_{22}) + 80\,(x_{31} + x_{32}) \le 1{,}050$
$1{,}200x_{12} + 1{,}400x_{22} + 800x_{32} \le 9{,}200$
$20x_{12} + 15x_{22} + 35x_{32} \le 250$
$100x_{12} + 60x_{22} + 80x_{32} \le 650$

and $x_{ij} \ge 0$ for all i, j

19. Let x_1, x_2, x_3 and x_4 = chair types 1, 2, 3 and 4 to be produced, respectively.

Max $Z = 12x_1 + 20x_2 + 18x_3 + 40x_4$

subject to $4x_1 + 9x_2 + 7x_3 + 10x_4 \le 6{,}000$
$x_1 + x_2 + 3x_3 + 40x_4 \le 4{,}000$

and $x_1, x_2, x_3, x_4 \ge 0$

Ans. $x_1 = 4{,}000/3$, $x_2 = x_3 = 0$, $x_4 = 200/3$ and Max Z = Rs 56,000/3.

21. Let x_1, x_2 and x_3 = cubic metric of earth processed at plants 1, 2 and 3, respectively.

Min $Z = 0.60x_1 + 0.36x_2 + 0.50x_3$

subject to

$0.58x_1 + 0.26x_2 + 0.21x_3 = 1{,}48{,}000$ (Stone count requirement)

$0.36x_1 + 0.22x_2 + 0.263x_3 \le 1{,}30.000$ (Carat requirement)

$x_1 \le 83{,}000$; $x_2 \le 3{,}10{,}000$;

$x_3 \le 1{,}90{,}000$ (Capacity requirement)

Ans. $x_1 = 61{,}700$; $x_2 = 3{,}10{,}000$; $x_3 = 1{,}50{,}500$ and Min Z = Rs 2,23,880.

4.5 SOME COMPLICATIONS AND THEIR RESOLUTION

In this section, some of the complications that may arise in applying the simplex method for solving both maximization and minimization LP problems and their resolution are discussed.

4.5.1 Unrestricted Variables

In actual practice, decision variables, x_j ($j = 1, 2, \ldots, n$) should have non-negative values. However, in many situations, one or more of these variables may have either positive, negative or zero value. Variables that can

assume positive, negative or zero value are called *unrestricted variables.* Since use of the simplex method requires that all the decision variables must have non-negative value at each iteration, therefore in order to convert an LP problem involving unrestricted variables into an equivalent LP problem having only non-negative variables, each of unrestricted variable is expressed as the difference of two non-negative variables.

Let variable x_r be unrestricted in sign. We define two new variables say x_r' and x_r'' such that

$$x_r = x_r' - x_r''; \qquad x_r', x_r'' \geq 0$$

If $x_r' \geq x_r''$, then $x_r \geq 0$, but and if $x_r' \leq x_r''$, then $x_r \leq 0$. Also if $x_r' = x_r''$, then $x_r = 0$. Hence depending on the value of x_r' and x_r'', the variable x_r can have either positive or negative sign. For example, the following LP problem

An unrestricted variable in an LP model can have either positive, negative or zero value.

$$\text{Maximize } Z = \sum_{j \neq r}^{n} c_j x_j + c_r x_r$$

subject to the constraints

$$\sum_{j \neq r}^{n} a_{ij} x_j + a_{ir} x_r = b_i, \quad i = 1, 2, \ldots, m$$

and $x_j \geq 0$, and x_r unrestricted in sign; $j = 1, 2, \ldots, n, \quad j \neq r$;

can be converted into its equivalent standard form as follows:

$$\text{Maximize } Z = \sum_{j \neq r}^{n} c_j x_j + c_r (x_r' + x_r'')$$

subject to the constraints

$$\sum_{j \neq r}^{n} a_{ij} x_j + a_{ir} (x_r' - x_r'') = b_i, \quad i = 1, 2, \ldots, m$$

and $x_j, x_r', x_r'' \geq 0; \quad j = 1, 2, \ldots, n, \quad j \neq r.$

Variables x_r' and x_r'' simultaneously cannot appear in the basis (since the column vectors corresponding to these variables are linearly dependent). Thus any of the following three cases may arise at the optimal solution:

(i) $x_r' = 0 \Rightarrow x_r = -x_r''$ (ii) $x_r'' = 0 \Rightarrow x_r = x_r'$ (iii) $x_r' = x_r'' = 0 \Rightarrow x_r = 0$

This indicates that the value of x_r' and x_r'' uniquely determines the values of the variable x_r.

Example 4.12 Use the simplex method to solve the following LP problem.

Maximize $Z = 3x_1 + 2x_2 + x_3$

subject to the constraints

(i) $2x_1 + 5x_2 + x_3 = 12$, (ii) $3x_1 + 4x_2 = 11$

and $x_2, x_3 \geq 0$, and x_1 unrestricted [*Bombay, BSc (Maths), 2001*]

Solution Introducing an artificial variable A_1 in the second constraint of the given LP problem in order to obtain the basis matrix as shown in Table 4.43. Since x_1 is unrestricted in sign, introducing the non-negative variables x_1' and x_2'' so that $x_1 = x_1' - x_1''$, where $x_1', \quad x_1'' \geq 0$. The standard form of the LP problem now becomes:

Maximize $Z = 3(x_1' - x_1'') + 2x_2 + x_3 - MA_1$

subject to the constraints

(i) $2(x_1' - x_1'') + 5x_2 + x_3 = 12$, (ii) $3(x_1' - x_1'') + 4x_2 + A_1 = 11$

and $x_1', x_1'', x_2, x_3, A_1 \geq 0$

The initial solution is shown in Table 4.43.

		$c_j \rightarrow$	3	– 3	2	1	– M	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	A_1	Min Ratio $\boldsymbol{x_B}/x_2$
1	x_3	12	2	– 2	(5)	1	0	12/5 →
– M	A_1	11	3	– 3	4	0	1	11/4
$Z = -11M + 12$		z_j	$-3M+2$	$3M-2$	$-4M+5$	1	– M	
		$c_j - z_j$	$3M+1$	$-3M-1$	$4M-3$ ↑	0	0	

Table 4.43 Initial Solution

In Table 4.43, $c_3 - z_3$ is the largest positive value, the non-basic variable x_2 is chosen to enter into the basis to replace basic variable x_3 in the basis. For this, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old)/5 (*key element*) $\quad R_2$ (new) $\rightarrow R_2$ (old) – $4R_1$ (new).

to get an improved solution shown in Table 4.44.

		$c_j \rightarrow$	3	– 3	2	1	– M	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	A_1	Min Ratio $\boldsymbol{x_B}/x_1'$
2	x_2	12/5	2/5	–2/5	1	1/5	0	$\frac{12}{5} \times \frac{5}{2} = 6$
– M	A_1	7/5	(7/5)	–7/5	0	– 4/5	1	$\frac{7}{5} \times \frac{5}{7} = 1$ →
$Z = -7M/5 + 24/5$		z_j	$-7M/5+4/5$	$7M/5-4/5$	2	$4M/5+2/5$	– M	
		$c_j - z_j$	$7M/5+11/5$ ↑	$-7M/5-11/5$	0	$-4M/5+3/5$	0	

Table 4.44 Improved Solution

In Table 4.44, as $c_1 - z_1$ in x'_1-column is positive, the non-basic variable x'_1 is chosen to enter into the basis to replace basic variable A_1 in the basis. For this, apply the following row operations:

R_2 (new) $\rightarrow R_2$ (old) × (5/7) (*key element*); $\quad R_1$ (new) $\rightarrow R_1$ (old) – (2/5) R_2 (new).

to get an improved solution shown in Table 4.45.

		$c_j \rightarrow$	3	– 3	2	1	
Basic Variables Coefficient	Basic Variables **B**	Basic Variables Value **b**(= $\boldsymbol{x_B}$)	x_1'	x_1''	x_2	x_3	Min Ratio $\boldsymbol{x_B}/x_3$
2	x_2	2	0	0	1	(3/7)	2/(3/7) = 14/3 →
3	x_1'	1	1	–1	0	– 4/7	—
$Z = 7$		z_j	3	– 3	2	– 6/7	
		$c_j - z_j$	0	0	0	13/7 ↑	

Table 4.45 Improved Solution

Further, in order to improve the solution given in Table 4.45, the non-basic variable x_3 is chosen to enter into the basis and remove basic variable x_2 from the basis. For this, apply the following row operations:

R_1 (new) $\rightarrow R_1$ (old) × (7/3) (*key element*); $\quad R_2$ (new) $\rightarrow R_2$ (old) + (4/7) R_1 (new)

to get an improved solution given in Table 4.46.

Table 4.46
Optimal Solution

		$c_j \rightarrow$	3	– 3	2	1
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1'	x_1''	x_2	x_3
1	x_3	14/3	0	0	7/3	1
3	x_1'	11/3	1	– 1	4/3	0
$Z = 47/3$		z_j	3	– 3	19/3	1
		$c_j - z_j$	0	0	– 13/3	0

Since in Table 4.46 all $c_j - z_j \leq 0$, an optimal solution: $x_1' = 11/3$ or $x_1 = x_1' - x_1'' = 11/3 - 0 = 11/3$; $x_3 = 14/3$ is arrived with Max $Z = 47/3$.

4.5.2 Tie for Entering Basic Variable (Key Column)

While solving an LP problem using simplex method two or more columns of simplex table may have same $c_j - z_j$ value (positive or negative depending upon the type of LP problem). In order to break this tie, the selection for key column (entering variable) can be made arbitrary. However, the number of iterations required to arrive at the optimal solution can be minimized by adopting the following rules:

(i) If there is a tie between two decision variables, then the selection can be made arbitrarily.

(ii) If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis.

(iii) If there is a tie between two slack (or surplus) variables, then the selection can be made arbitrarily.

4.5.3 Tie for Leaving Basic Variable (Key Row) – Degeneracy

While solving an LP problem a situation may arise where either the minimum ratio to identify the basic variable to leave the basis is not unique or value of one or more basic variables in the $\boldsymbol{x_B}$ becomes zero. This causes the problem of degeneracy.

Usually, the problem of degeneracy arises due to redundant constraint, i.e. one or more of the constraints of LP problem are not part of feasible solution space. For example, constraints such as $x_1 \leq 5$, $x_2 \leq 5$ and $x_1 + x_2 \leq 5$ in the LP problem make constraint $x_1 + x_2 \leq 5$ unnecessary (redundant).

Degeneracy may occur at any iteration of the simplex method. In order to break the tie in the minimum ratios, the selection can be made arbitrarily. However, the number of iterations required to arrive at the optimal solution can be minimized by adopting the following rules.

(i) Divide the coefficients of slack variables in the simplex table where degeneracy is seen by the corresponding positive numbers of the key column in the row, starting from left to right.

(ii) Compare the ratios in step (i) from left to right columnwise, select the row that contains the smallest ratio.

Remark When there is a tie between a slack and artificial variable to leave the basis, preference should be given to the artificial variable for leaving the basis.

Example 4.13 Solve the following LP problem

Maximize $Z = 3x_1 + 9x_2$

subject to the constraints

(i) $x_1 + 4x_2 \leq 8$, (ii) $x_1 + 2x_2 \leq 4$

and $x_1, x_2 \geq 0$

Solution Adding slack variables s_1 and s_2 to the constraints, the problem can be expressed as

Maximize $Z = 3x_1 + 9x_2 + 0s_2 + 0s_2$

subject to the constraints

(i) $x_1 + 4x_2 + s_1 = 8$, (ii) $x_1 + 2x_2 + s_2 = 4$

and $x_1, x_2, s_1, s_2 \geq 0$

The initial basic feasible solution is given in Table 4.47. Since $c_2 - z_2 = 9$ is the largest positive value, therefore, variable x_2 is selected to be entered into the basis. However, both variables s_1 and s_2 are eligible to leave the basis as the minimum ratio is same, i.e. 2, so there is a tie among the ratio in rows s_1 and s_2. This causes the problem of degeneracy. To obtain the key row for resolving degeneracy, apply the following procedure:

(i) Write the coefficients of the slack variables as shown in Table 4.47.

		$c_j \rightarrow$	3	9	0	0	
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	*Min Ratio* $\boldsymbol{x_B}/x_2$
0	s_1	8	1	4	1	0	8/4 = 2
0	s_2	4	1	(2)	0	1	4/2 = 2
$Z = 0$		z_j	0	0	0	0	
		$c_j - z_j$	3	9 ↑	0	0	

Table 4.47 Initial Solution

Row	x_2-*column (Key Column)*	*Column* s_1	*Column* s_2
s_1	4	1	0
s_2	2	0	1

(ii) Dividing the coefficients of slack variables s_1 and s_2 by the corresponding elements in the key column as shown below in the table.

Row	x_2-*column (Key Column)*	*Column* s_1	*Column* s_2
s_1	4	1/4 = 1/4	0/4 = 0
s_2	2	0/2 = 0	1/2 = 1/2

(iii) Comparing the ratios of Step (ii) from left to right columnwise, the minimum ratio (i.e., 0/2 = 0) occurs in the s_2-row. Thus, variable s_2 is selected to leave the basis. The new solution is shown in Table 4.48.

		$c_j \rightarrow$	3	9	0	0
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x}_B)$	x_1	x_2	s_1	s_2
0	s_1	0	− 1	0	1	− 2
9	x_2	2	1/2	1	0	1/2
$Z = 18$		z_j	9/2	9	0	9/2
		$c_j - z_j$	− 3/2	0	0	− 9/2

Table 4.48 Optimal Solution

In Table 4.48, all $c_j - z_j \leq 0$. Hence, an optimal solution is arrived at. The optimal basic feasible solution is: $x_1 = 0$, $x_2 = 2$ and Max $Z = 18$.

4.6 TYPES OF LINEAR PROGRAMMING SOLUTIONS

While solving any LP problem using simplex method, at the stage of optimal solution, the following three types of solutions may exist:

4.6.1 Alternative (Multiple) Optimal Solutions

The $c_j - z_j$ values in the simplex table indicates the contribution in the objective function value by each unit of a variable chosen to enter into the basis. Also, an optimal solution to a maximization LP problem is reached when all $c_j - z_j \leq 0$. But, if $c_j - z_j = 0$ for a non-basic variable column in the optimal simplex table and such non-basic variable is chosen to enter into the basis, then another optimal solution so obtained will show no improvement in the value of objective function.

Alternative optimal solutions arise when $c_j - z_j = 0$ for non-basic variable columns in the simplex table.

Example 4.14 Solve the following LP problem.

Maximize $Z = 6x_1 + 4x_2$

subject to the constraints

(i) $2x_1 + 3x_2 \leq 30$, (ii) $3x_1 + 2x_2 \leq 24$, (iii) $x_1 + x_2 \geq 3$

and $x_1, x_2 \geq 0$.

Solution Adding slack variables s_1, s_2, surplus variable s_3 and artificial variable A_1 in the constraint set, the LP problem becomes

Maximize $Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $2x_1 + 3x_2 + s_1 = 30$, (ii) $3x_1 + 2x_2 + s_2 = 24$ (iii) $x_1 + x_2 - s_3 + A_1 = 3$

and $x_1, x_2, s_1, s_2, s_3, A_1 \geq 0$

The optimal solution: $x_1 = 8$, $x_2 = 0$ and Max $Z = 48$ for this LP problem is shown in Table 4.49.

Table 4.49 Optimal Solution

		$c_j \rightarrow$	6	4	0	0	0	
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3	*Min Ratio* $\boldsymbol{x_B}/x_2$
0	s_1	14	0	(5/3)	1	− 2/3	0	14/(15/3) = 42/5 →
0	s_3	5	0	− 1/3	0	1/3	1	—
6	x_1	8	1	2/3	0	1/3	0	8/(2/3) = 12
Z = 48		z_j	6	4	0	2	0	
		$c_j - z_j$	0	0 ↑	0	− 2	0	

In Table 4.49, $c_2 - z_2 = 0$ corresponds to a non-basic variable, x_2. Thus, an alternative optimal solution can also be obtained by entering variable x_2 into the basis and removing basic variable, s_1 from the basis. The new solution is shown in Table 4.50.

Table 4.50 Alternative Solution

		$c_j \rightarrow$	6	4	0	0	0
Basic Variables Coefficient	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Value* $\boldsymbol{b}(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3
4	x_2	42/5	0	1	3/5	− 2/5	0
0	s_3	39/5	0	0	1/5	1/5	1
6	x_1	12/5	1	0	− 2/5	3/5	0
Z = 48		z_j	6	4	0	2	0
		$c_j - z_j$	0	0	0	− 2	0

The optimal solution shown in Table 4.50 is: $x_1 = 12/5$, $x_2 = 42/5$ and Max $Z = 48$. Since this optimal solution shows no change in the value of objective function, it is an alternative solution.

Once again, $c_3 - z_3 = 0$ corresponds to non-basic variable, s_1. This again indicates that an alternative optimal solution exists. The infinite number of solutions that can be obtained for this LP problem are as follows:

Variables	*Solution Values*		*General Solution*
	1	*2*	
x_1	8	12/5	$x_1 = 8\lambda + (12/5)(1-\lambda)$
x_2	0	42/5	$x_2 = 0\lambda + (42/5)(1-\lambda)$
s_1	14	0	$s_1 = 14\lambda + (0)(1-\lambda)$
s_3	5	39/5	$s_3 = 5\lambda + (39/5)(1-\lambda)$

For each arbitrary value of λ, the value of objective function will remain same.

4.6.2 Unbounded Solution

In a maximization LP problem, if $c_j - z_j > 0$ ($c_j - z_j < 0$ for a minimization case) corresponds to a non-basic variable column in simplex table, and all a_{ij} values in this column are negative, then minimum ratio to decide basic variable to leave the basis can not be calculated. It is because negative value in denominator would indicate the entry of a non-basic variable in the basis with a negative value (an infeasible solution). Also, a zero value in the denominator would result in a ratio having an infinite value and would indicate that the value of non-basic variable could be increased infinitely with any of the current basic variables being removed from the basis.

Example 4.15 Solve the following LP problem.

Maximize $Z = 3x_1 + 5x_2$

subject to the constraints

(i) $x_1 - 2x_2 \le 6$, (ii) $x_1 \le 10$, (iii) $x_2 \ge 1$

and $x_1, x_2 \ge 0$

Solution Adding slack variables s_1, s_2, surplus variable s_3 and artificial variable A_1 in the constraint set. Then the standard form of LP problem becomes

Maximize $Z = 3x_1 + 5x_2 + 0s_1 + 0s_2 + 0s_3 - MA_1$

subject to the constraints

(i) $x_1 - 2x_2 + s_1 = 6$, (ii) $x_1 + s_2 = 10$, (iii) $x_2 - s_3 + A_1 = 1$

and $x_1, x_2, s_1, s_2, s_3, A_1 \ge 0$

The initial solution to this LP problem is shown in Table 4.51.

Unbounded solution occurs when value of decision variables in the solution of LP problem becomes infinitely large without violating any given constraints.

		$c_j \rightarrow$	*3*	*5*	*0*	*0*	*0*	*−M*	
Basic Variables Coefficient	*Basic Variables* **B**	*Basic Variables Value* **b** (= x_B)	x_1	x_2	s_1	s_2	s_3	A_1	*Min Ratio* x_B/x_2
0	s_1	6	1	−2	1	0	0	0	–
0	s_2	10	1	0	0	1	0	0	–
$-M$	A_1	1	0	(1)	0	0	−1	1	1 →
$Z = -M$		z_j	0	$-M$	0	0	M	$-M$	
		$c_j - z_j$	3	$5 + M$ ↑	0	0	$-M$	0	

Table 4.51 Initial Solution

Iteration 1: Since $c_2 - z_2 \ge 0$, non-basic variable x_2 is chosen to enter into the basis in place of basic variable A_1. The new solution is shown in Table 4.52.

		$c_j \rightarrow$	3	5	0	0	0	$-M$
Basic Variables Coefficient	Basic Variables $\boldsymbol{B}$	Basic Variables Value $\boldsymbol{b}$ $(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	s_3	A_1
0	s_1	8	1	0	1	0	-2	2
0	s_2	10	1	0	0	1	0	0
5	x_2	1	0	1	0	0	-1	1
$Z = 5$		z_j	0	5	0	0	-5	5
		$c_j - z_j$	3	0	0	0	5	$-M-5$

Table 4.52

In Table 4.52, $c_5 - z_5 = 5$ is largest positive, so variable s_3 should enter into the basis. But, coefficients in the 's_3' column are all negative or zero. This indicates that s_3 cannot be entered into the basis. However, the value of s_3 can be increased infinitely without removing any one of the basic variables. Further, since s_3 is associated with x_2 in the third constraint, x_2 will also be increased infinitely because it can be expressed as $x_2 = 1 + s_3 - A_1$. Hence, the solution to the given problem is unbounded.

4.6.3 Infeasible Solution

If LP problem solution does not satisfy all of the constraints, then such a solution is called *infeasible solution*. Also, infeasible solution occurs when all $c_j - z_j$ values satisfy optimal solution condition but at least one of artificial variables appears in the basis with a positive value. This situation may occur when an LP model is either improperly formulated or more than two of the constraints are incompatible.

Example 4.16 Solve the following LP problem

Maximize $Z = 6x_1 + 4x_2$

subject to the constraints

(i) $x_1 + x_2 \leq 5$, (ii) $x_2 \geq 8$

and $x_1, x_2 \geq 0$.

Solution Adding slack, surplus and artificial variables, the standard form of LP problem becomes

Maximize $Z = 6x_1 + 4x_2 + 0s_1 + 0s_2 - MA_1$

subject to the constraints

(i) $x_1 + x_2 + s_1 = 5$, (ii) $x_2 - s_2 + A_1 = 8$

and $x_1, x_2, s_1, s_2, A_1 \geq 0$

The initial solution to this LP problem is shown in Table 4.53.

		$c_j \rightarrow$	6	4	0	0	$-M$	
Basic Variables Coefficient	Basic Variables $\boldsymbol{B}$	Basic Variables Value $\boldsymbol{b}$ $(= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	A_1	Min Ratio $\boldsymbol{x_B}/x_2$
6	s_1	5	1	(1)	1	0	0	$5/1 \rightarrow$
$-M$	A_1	8	0	1	0	-1	1	8/1
$Z = 30 - 8M$		z_j	6	$6 - M$	0	M	$-M$	
		$c_j - z_j$	0	$-2 + M$ ↑	0	$-M$	0	

Table 4.53
Initial Solution

Iteration 1: Since $c_2 - z_2 = M - 2 (\geq 0)$, non-basic variable x_2 is chosen to enter into the basis to replace basic variable x_1. The new solution is shown in Table 4.58.

		$c_j \rightarrow$	6	4	0	0	$-M$
Basic Variables Coefficients	*Basic Variables* $\boldsymbol{B}$	*Basic Variables Values* $\boldsymbol{b}\ (= \boldsymbol{x_B})$	x_1	x_2	s_1	s_2	A_1
4	x_2	5	1	1	1	0	0
$-M$	A_1	3	-1	0	-1	-1	1
$Z = 20 - 3M$		z_j	$4 + M$	4	$4 + M$	M	$-M$
		$c_j - z_j$	$2 - M$	0	$-4 - M$	$-M$	0

Table 4.54 Optimal but Infeasible Solution

In table 5.44, since all $c_j - z_j \leq 0$, the current solution is optimal. But this solution is not feasible for the given LP problem because values of decision variables are: $x_1 = 0$ and $x_2 = 5$ violates second constraint, $x_2 \geq 8$. The presence of artificial variable $A_1 = 3$ in the solution also indicates that the optimal solution violates the second constraint ($x_2 \geq 8$) by 3 units.

CONCEPTUAL QUESTIONS

1. Define slack and surplus variables in a linear programming problem.
2. Explain the various steps of the simplex method involved in the computation of an optimum solution to a linear programming problem.
3. (a) Give outlines of the simplex method in linear programming.
   (b) What is simplex? Describe the simplex method of solving linear programming problem.
   (c) What do you understand by the term two-phase method of solving linear programming problem?
   (d) Outline the simplex method in linear programming. Why is it called so?
   (e) Explain the purpose and procedure of the simplex method.
4. What do you mean by an optimal basic feasible solution to a linear programming problem?
5. Given a general linear programming problem, explain how you would test whether a basic feasible solution is an optimal solution or not. How would you proceed to change the basic feasible solution in case it is not optimal?
6. Explain the meaning of basic feasible solution and degenerate solution in a linear programming problem.
7. Explain what is meant by the terms degeneracy and cycling in linear programming? How can these problems be resolved?
8. Explain the term artificial variables and its use in linear programming.
9. What are artificial variables? Why do we need them? Describe the two-phase method of solving an LP problem with artificial variables.
10. What is the significance of $c_j - z_j$ numbers in the simplex table? Interpret their economic significance in terms of marginal worth.
11. What is meant by the term opportunity cost? How is this concept used in designing a test for optimality?
12. How do the graphical and simplex methods of solving LP problems differ from each other? In what ways are they same?
13. How do maximization and minimization problems differ when applying the simplex method?
14. What is the reason behind the use of the minimum ratio test in selecting the key row? What might happen without it?
15. What conditions must exist in a simplex table to establish the existence of an alternative solution? No feasible solution? Unbounded solution? Degeneracy?

SELF PRACTICE PROBLEMS B

1. Solve the following LP problems using the simplex method.

(i) Max $Z = 3x_1 + 2x_2$
subject to $x_1 + x_2 \leq 4$
$x_1 - x_2 \leq 2$
and $x_1, x_2 \geq 0$

(ii) Max $Z = 5x_1 + 3x_2$
subject to $x_1 + x_2 \leq 2$
$5x_1 + 2x_2 \leq 10$
$3x_1 + 8x_2 \leq 12$
and $x_1, x_2 \geq 0$

(iii) Max $Z = 3x_1 + 2x_2 + 5x_3$
subject to $x_1 + 2x_2 + x_3 \leq 430$
$3x_1 + 2x_3 \leq 460$
$x_1 + 4x_3 \leq 420$
and $x_1, x_2, x_3 \geq 0$
[*Shivaji, MSc* (*Maths*), *1995*]

(iv) Max $Z = x_1 - 3x_2 + 2x_3$
subject to $3x_1 - x_2 + 3x_3 \leq 7$
$-2x_1 + 4x_2 \leq 12$
$-4x_1 + 3x_2 + 8x_3 \leq 10$
and $x_1, x_2, x_3 \geq 0$
[*Meerut, MSc* (*Maths*), *1997; Bhathiar MSc* (*Maths*), *1996*]

(v) Max $Z = 4x_1 + 5x_2 + 9x_3 + 11x_4$
subject to
$x_1 + x_2 + x_3 + x_4 \leq 15$
$7x_1 + 5x_2 + 3x_3 + 2x_4 \leq 120$
$3x_1 + 5x_2 + 10x_3 + 15x_4 \leq 100$
and $x_1, x_2, x_3, x_4 \geq 0$

(vi) Max $Z = x_1 + x_2 + x_3$
subject to $4x_1 + 5x_2 + 3x_3 \leq 15$
$10x_1 + 7x_2 + x_3 \leq 12$
and $x_1, x_2, x_3 \geq 0$

(vii) Max $Z = x_1 + x_2 + x_3$
subject to $3x_1 + 2x_2 + x_3 \le 3$
$2x_1 + x_2 + 2x_3 \le 2$
and $x_1, x_2, x_3 \ge 0$

(viii) Max $Z = 2x_1 + 4x_2 + 3x + x_4$
subject to $x_1 + 3x_2 + x_4 \le 4$
$2x_1 + x_2 \le 3$
$x_2 + 4x_3 + x_4 \le 3$
and $x_1, x_2, x_3, x_4 \ge 0$

(ix) Max $Z = 107x_1 + x_2 + 2x_3$
subject to $14x_1 + x_2 - 6x_3 + 3x_4 \le 7$
$16x_1 + 0.5x_2 + 6x_3 \le 5$
$3x_1 - x_2 - x_3 \le 10$
and $x_1, x_2, x_3, x_4 \ge 0$
[*Sambalpur, MSc* (*Maths*), *2003*]

(x) Max $Z = 4x_1 + 10x_2$
subject to $2x_1 + x_2 \le 50$
$2x_1 + 5x_2 \le 100$
$2x_1 + 3x_2 \le 90$
and $x_1, x_2 \ge 0$

(xi) Max $Z = x_1 - x_2 + 3x_3$
subject to $x_1 + x_2 + x_3 \le 10$
$2x_1 - x_3 \le 2$
$2x_1 - 2x_2 + 3x_3 \le 0$
and $x_1, x_2 \ge 0$
[*Karnataka, MSc* (*Maths*), *2004*]

(xii) Max $Z = 4x_1 + x_2 + 3x_3 + 5x_4$
subject to
$4x_1 - 6x_2 - 5x_3 + 4x_4 \ge -20$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$8x_1 - 3x_2 + 3x_3 + 2x_4 \le 20$
and $x_1, x_2, x_3, x_4 \ge 0$
has an unbound solution.

2. Use the two-phase method to solve the following LP problems:

(i) Max $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 2$
$x_2 \le 4$
and $x_1, x_2 \ge 0$

(ii) Min $Z = 3x_1 - x_2 - x_3$
subject to $x_1 - 2x_2 + x_3 \le 11$
$-4x_1 + x_2 + 2x_3 \ge 3$
$-2x_1 + x_3 = 1$
and $x_1, x_2, x_3 \ge 0$

(iii) Min $Z = 7.5x_1 - 3x_2$
subject to $3x_1 - x_2 - x_3 \ge 3$
$x_1 - x_2 + x_3 \ge 2$
and $x_1, x_2, x_3 \ge 0$

(iv) Min $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 2$
$x_2 \le 4$
and $x_1, x_2 \ge 0$

(v) Min $Z = 5x_1 + 8x_2$
subject to $3x_1 + 2x_2 \ge 3$
$x_1 + 4x_2 \ge 4$
$x_1 + x_2 \le 5$
and $x_1, x_2 \ge 0$

(vi) Min $Z = 3x_1 + 2x_2$
subject to $2x_1 + x_2 \ge 2$
$3x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

3. Use penalty (Big-M) method to solve the following LP problems:

(i) Min $Z = 3x_1 - x_2$
subject to $2x_1 + x_2 \ge 2$
$x_1 + 3x_2 \le 3$
$x_2 \ge 4$
$x_1, x_2 \ge 0$

(ii) Min $Z = 2x_1 + x_2$
subject to $3x_1 + x_2 = 3$
$4x_1 + 3x_2 \ge 6$
$x_1 + 2x_2 \le 4$
$x_1, x_2 \ge 0$

(iii) Max $Z = 3x_1 + 2x_2 + 3x_3$
subject to $2x_1 + x_2 + x_3 \le 2$
$3x_1 + 4x_2 + 2x_3 \le 8$
and $x_1, x_2, x_3 \ge 0$

(iv) Max $Z = x_1 + x_2 + x_4$
subject to $x_1 + x_2 + x_3 + x_4 = 4$
$x_1 + 2x_2 + x_3 + x_4 = 4$
$x_1 + 2x_2 + x_3 = 4$

(v) Min $Z = x_1 - 3x_2 + 2x_3$
subject to $3x_1 - x_2 + 2x_3 \le 7$
$-2x_1 + 4x_2 \le 12$
$-4x_1 + 3x_2 + 8x_3 \le 10$
and $x_1, x_2, x_3 \ge 0$
[*Meerut MSc* (*Maths*), *2005*]

(vi) Min $Z = 5x_1 + 2x_2 + 10x_3$
subject to $x_1 - x_3 \le 10$
$x_2 + x_3 \ge 10$
and $x_1, x_2, x_3 \ge 0$

4. Solve the following LP problems and remove the complication (if any).

(i) Max $Z = 2x_1 + 3x_2 + 10x_3$
subject to $x_1 + 2x_3 = 2$
$x_2 + x_3 = 1$
and $x_1, x_2, x_3 \ge 0$

(ii) Max $Z = 5x_1 - 2x_2 + 3x_3$
subject to $2x_1 + 2x_2 - x_3 \ge 2$
$3x_1 - 4x_2 \le 3$
$x_2 + 3x_3 \le 5$
and $x_1, x_2, x_3 \ge 0$

(iii) Max $Z = 5x_1 + 3x_2$
subect to $x_1 + x_2 \le 2$
$5x_1 + 2x_2 \le 10$
$3x_1 + 8x_2 \le 12$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 22x_1 + 30x_2 + 25x_3$
subject to $2x_1 + 2x_2 + \le 100$
$2x_1 + x_2 + x_3 \ge 100$
$x_1 + 2x_2 + 2x_3 \le 100$
and $x_1, x_2, x_3 \ge 0$

(v) Max $Z = 8x_2$
subject to $x_1 - x_2 \ge 0$
$2x_1 + 3x_2 \le -6$
and x_1, x_2 unrestricted.

5. Solve the following LP problems to show that these have alternative optimal solutions.

(i) Max $Z = 6x_1 + 3x_2$
subject to $2x_1 + x_2 \le 8$
$3x_1 + 3x_2 \le 18$
$x_2 \le 3$
and $x_1, x_2 \ge 0$

(ii) Min $Z = 2x_1 + 8x_2$
subject to $5x_1 + x_2 \ge 10$
$2x_1 + 2x_2 \ge 14$
$x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

(iii) Max $Z = x_1 + 2x_2 + 3x_3 - x_4$
subject to $x_1 + 2x_2 + 3x_3 = 15$
$2x_1 + x_2 + 5x_3 \ge 20$
$x_1 + x_2 + x_3 + x_4 \ge 10$
and $x_1, x_2, x_3, x_4 \ge 0$

6. Solve the following LP problems to show that these have an unbounded solution.

(i) Max $Z = -2x_1 + 3x_2$
subject to $x_1 \le 5$
$2x_1 - 3x_2 \le 6$
and $x_1, x_2 \ge 0$

(ii) Max $Z = 3x_1 + 6x_2$
subject to $3x_1 + 4x_2 \ge 12$
$-2x_1 + x_2 \le 4$
and $x_1, x_2 \ge 0$

(iii) Max $Z = 107x_1 + x_2 + 2x_3$
subject to $14x_1 + x_2 - 6x_3 + 3x_4 = 7$
$16x_1 + 0.5x_2 - 6x_3 \le 5$
$3x_1 - x_2 - x_3 \le 0$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 6x_1 - 2x_2$
subject to $2x_1 - x_2 \le 2$
$x_1 \le 4$
and $x_1, x_2, x_3, x_4 \ge 0$

7. Solve the following LP problems to show that these have no feasible solution.

(i) Max $Z = 2x_1 + 3x_2$
subject to $x_1 - x_2 \ge 4$
$x_1 + x_2 \le 6$
$x_1 \le 2$
and $x_1, x_2 \ge 0$

(ii) Max $Z = 4x_1 + x_2 + 4x_3 + 5x_4$
subject to $4x_1 + 6x_2 - 5x_3 + 4x_4 \ge -20$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$3x_1 - 2x_2 + 4x_3 + x_4 \le 10$
$8x_1 - 3x_2 - 3x_3 + 2x_4 \le 20$
and $x_1, x_2, x_3, x_4 \ge 0$

(iii) Max $Z = x_1 + 3x_2$
subject to $x_1 - x_2 \ge 1$
$3x_1 - x_2 \le -3$
and $x_1, x_2 \ge 0$

(iv) Max $Z = 3x_1 + 2x_2$
subject to $2x_1 + x_2 \le 2$
$3x_1 + 4x_2 \ge 12$
and $x_1, x_2 \ge 0$

[*Meerut, MSc* (*Maths*), *2006*]

HINTS AND ANSWERS

1. (i) $x_1 = 3$, $x_2 = 1$ and Max $Z = 11$
(ii) $x_1 = 2$, $x_2 = 0$ and Max $Z = 10$
(iii) $x_1 = 0$, $x_2 = 100$, $x_3 = 230$ and Max $Z = 1{,}350$
(iv) Max $Z^* = -x_1 + 3x_2 - 2x_3$ where $Z^* = -Z$
$x_1 = 4, x_2 = 5, x_3 = 0$ and $Z^* = 11$.
(v) $x_1 = 50/7, x_2 = 0, x_3 = 55/7$ and Max $Z = 695/7$
(vi) $x_1 = 0$, $x_2 = 0$, $x_3 = 5$ and Max $Z = 5$
(vii) $x_1 = 0$, $x_2 = 0$, $x_3 = 1$ and Max $Z = 3$
(viii) $x_1 = 1$, $x_2 = 1$, $x_3 = 1/2$ and Max $Z = 13/2$
(ix) Divide the first equation by 3 (coefficient of x_4)
(x) $x_1 = 0$, $x_2 = 0$ and Max $Z = 200$
(xi) $x_1 = 0$, $x_2 = 6$, $x_3 = 4$ and Max $Z = 6$

2. (i) $x_1 = 1$, $x_2 = 0$, and Max $Z = 6$
(ii) All $c_j - z_j \le 0$ but $Z = -5/4$ (< 0) and artificial variable $A_1 = 5/4$ appears in the basis with positive value. Thus the given LP problem has no feasible solution.
(iii) $x_1 = 5/4$, $x_2 = 0$, $x_3 = 0$ and Min $Z = 75/8$
(iv) $x_1 = 2$, $x_2 = 0$ and Max $Z = 6$
(v) $x_1 = 0$, $x_2 = 5$ and Max $Z = 40$
(vi) All $c_j - z_j \ge 0$, but $Z = -4$ (< 0) and artificial variable $A_1 = 4$ appears in the basis with a positive value. Thus the given LP problem has no feasible solution.

3. (i) $x_1 = 3$, $x_2 = 0$ and Max $Z = 9$
(ii) $x_1 = 3/5$, $x_2 = 6/5$ and Min $Z = 12/5$
(iii) All $c_j - z_j \ge 0$ artificial variable $A_1 = 0$ appears in the basis with zero value. Thus an optimal solution to the given LP problem exists.
(iv) Introduce artificial variable only in the third constraint.
$x_1 = 0, x_2 = 0, x_3 = 0, x_5 = 0$ and Max $Z = 4$.
(v) $x_1 = 4, x_2 = 5$ and Min $Z = -11$
(vi) $x_1 = 0, x_2 = 10, x_3 = 0$ and Min $Z = 20$

4. (i) $x_1 = 0$, $x_2 = 1$, $x_3 = 0$ and Max $Z = 3$
(ii) Degeneracy occurs at the initial stage. One of the variable eligible to leave the basis is artificial variable, therefore, there is no need of resolving degeneracy. Remove the artificial variable from the basis.
$x_1 = 23/3$, $x_2 = 5$, $x_3 = 0$ and Max $Z = 85/3$
(iii) $x_1 = 2$, $x_2 = 0$ and Max $Z = 10$
(iv) $x_1 = 100/3$, $x_2 = 50/3$, $x_3 = 50/3$ and Max $Z = 1{,}650$
(v) $x_1'' = 6/5$ or $x_1 = -6/5$ or $x_2'' = 6/5$ or $x_2 = -6/5$ and Max $Z = -48/5$.

5. (i) (a) $x_1 = 4$, $x_2 = 0$ and Max $Z = 24$
(b) $x_1 = 5/2$, $x_2 = 3$, and Max $Z = 24$
(ii) (a) $x_1 = 32/6$, $x_2 = 10/6$ and Min $Z = 24$
(b) $x_1 = 12$, $x_2 = 0$ and Min $Z = 24$

6. (i) At the current solution: $x_1 = 5$, $x_2 = 9$ and Max $Z = 15$, it may be observed that $c_2 - z_2 = 3/2$ but all elements in the second column are negative. Solution is unbounded.

(ii) At the current solution: $x_2 = 4$, $s_1 = 4$ and Max $Z = 24$, it may be observed that $c_2 - z_2 = 15$ but all elements in the second column are negative. Solution is unbounded.

(iii) At second best solution, $c_3 - z_3 = 113/3$ but all elements in the third column are negative. Solution is unbounded; $x_1 = 0$, $x_4 = 7/3$, $s_1 = 5$ and Max $Z = 0$.

(iv) Optimal solution: $x_1 = 4$, $x_2 = 6$ and Max $Z = 12$. Since in the initial simplex table all the elements are negative in the second column, the feasible solution is unbounded but the optimal solution is bounded.

7. (i) $x_1 = 2, x_2 = 0, A_1 = 2$ and M Max $Z = 9 - 4M$; Infeasible solution.

(iv) $x_1 = 0, x_2 = 2, A_1 = 2$ and Max $Z = 4 - 4M$; Infeasible solution.

CHAPTER SUMMARY

The simplex method is an interactive procedure for reaching the optimal solution to any LP problem. It consists of a series of rules that, in effect, algebraically examine corner (extreme) points of the solution space in a systematic way. Each step moves towards the optimal solution by increasing profit or decreasing cost, while maintaining feasibility. The simplex method consists of five steps: (i) identifying the pivot column, (ii) identifying the pivot row and key element, (iii) replacing the pivot row, (iv) computing new values for each remaining row; and (v) computing the z_j and $c_j - z_j$ values and examining for optimality. Each step of this iterative procedure is displayed and explained in a simplex table both for maximization and minimization LP problems.

CHAPTER CONCEPTS QUIZ

True or False

1. The major difference between slack and artificial variables is that an artificial can never be zero.
2. If an optimal solution is degenerate, then there are alternate optimal solutions of the LP problem.
3. An infeasible solution is characterized as one where one constraint is violated.
4. If the objective function coefficient in the c_j row above an artificial variable is – M, then the problem is a minimization problem.
5. In the simplex method, the initial solution contains only slack variable in the product mix.
6. If there is a tie between decision variable and a slack (or surplus) variable, then select the decision variable to enter into the first basis.
7. An optimal solution to the maximized LP problem is reached if all $c_j - z_j \geq 0$.
8. Variables which can assume negative, positive or zero value are called unrestricted variables.
9. All the rules and procedures of the simplex method are identical whether solving a maximization or minimization LP problem.
10. Artificial variables are added to a linear programming problem to aid in the finding an optimal solution.

Fill in the Blanks

11. In the simplex method, the ———— column contains the variables which are currently in the solution; the values of these variables can be read from the ———— column.
12. A ———— variable represents amounts by which solution values exceed a resource.
13. The simplex method examines the ———— points in a systematic manner, repeating the same set of steps of the algorithm until an ———— solution is reached.
14. ———— occurs when there is no solution that satisfies all of the constraints in the linear programming problem.
15. The value for the replacing row must be ———— before computing the values for the ———— rows.
16. Entries in the $c_j - z_j$ rows are known as ———— costs.
17. Optimality is indicated for a maximization problem when all elements in the $c_j - z_j$ rows are ————, while for a minimization problem all elements must be ————.
18. In Big-M method, ———— basic feasible solution is obtained by assigning ———— value to the original value.
19. In the two-phase method, an ———— variable is never considered for re-entry into the basis.
20. ———— occurs when there is no finite solution in the LP problem.

Multiple Choice

21. The role of artificial variables in the simplex method is
    (a) to aid in finding an initial solution
    (b) to find optimal dual prices in the final simplex table
    (c) to start phases of simplex method
    (d) all of the above
22. For a maximization problem, the objective function coefficient for an artificial variable is
    (a) + M (b) – M
    (c) Zero (d) none of the above
23. If a negative value appears in the solution values ($\mathbf{x_B}$) column of the simplex table, then
    (a) the solution is optimal
    (b) the solution is infeasible
    (c) the solution is unbounded
    (d) all of the above
24. At every iteration of the simplex method, for minimization problem, a variable in the current basis is replaced with another variable that has
    (a) a positive $c_j - z_j$ value
    (b) a negative $c_j - z_j$ value
    (c) $c_j - z_j = 0$
    (d) none of the above

25. In the optimal simplex table, $c_j - z_j = 0$ value indicates
    (a) unbounded solution (b) cycling
    (c) alternative solution (d) infeasible solution
26. For maximization LP model, the simplex method is terminated when all values
    (a) $c_j - z_j \leq 0$ (b) $c_j - z_j \geq 0$
    (c) $c_j - z_j = 0$ (d) $z_j \leq 0$
27. A variable which does not appear in the basic variable (***B***) column of simplex table is
    (a) never equal to zero (b) always equal to zero
    (c) called a basic variable (d) none of the above
28. If for a given solution a slack variable is equal to zero then
    (a) the solution is optimal (b) the solution is infeasible
    (c) the entire amount of resource with the constraint in which the slack variable appears has been consumed.
    (d) all of the above
29. If an optimal solution is degenerate, then
    (a) there are alternative optimal solutions
    (b) the solution is infeasible
    (c) the solution is of no use to the decision-maker
    (d) none of the above
30. To formulate a problem for solution by the simplex method, we must add artificial variable to
    (a) only equality constraints
    (b) only 'greater than' constraints
    (c) both (a) and (b)
    (d) none of the above
31. If any value in x_B – Column of final simplex table is negative, then the solution is
    (a) unbounded (b) infeasible
    (c) optimal (d) none of the above
32. If all a_{ij} values in the incoming variable column of the simplex table are negative, then
    (a) solution is unbounded (b) there are multiple solutions
    (c) there exist no solution (c) the solution is degenerate
33. If an artificial variable is present in the 'basic variable' column of optimal simplex table, then the solution is
    (a) infeasible (b) unbounded
    (c) degenerate (d) none of the above
34. The per unit improvement in the solution of the minimization LP problem is indicated by the negative value of
    (a) $c_j - z_j$ in the decision-variable column
    (b) $c_j - z_j$ in the slack variable column
    (c) $c_j - z_j$ in the surplus variable column
    (d) none of the above
35. To convert ≥ inequality constraints into equality constraints, we must
    (a) add a surplus variable
    (b) subtract an artificial variable
    (c) substract a surplus variable and an artificial variable
    (d) add a surplus variable and substract an artificial variable

Answers to Quiz

1. F	2. F	3. T	4. F	5. F	6. T	7. T	8. T	9. F	10. T
11. product mix; quantity		12. Suplus		13. Extreme; optimal		14. Infeasibility		15. Computed; remaining	
16. reduced		17. non positive, non negative				18. Initial; zero		19. Artificial	20. Unboundedness
21. (d)	22. (b)	23. (b)	24. (b)	25. (c)	26. (a)	27. (b)	28. (c)	29. (d)	30. (c)
31. (b)	32. (a)	33. (a)	34. (d)	35. (c)					

CASE STUDY

Case 4.1: New Bharat Bank

New Bharat Bank is planning its operations for the next year. The bank makes five types of loans. These are listed below, together with the annual return (in per cent);

Type of Loan	*Annual Return (per cent)*
Education	15
Furniture	12
Automobile	9
Home Construction	10
Home repair	7

Legal requirements and bank policy place the following limits on the amounts of the various types of loans: Education loans cannot exceed 10 per cent of the total amount of loans. The amount of education and furniture loans together cannot exceed 20 per cent of the total amount of loans. Home construction loan must be at least 40 per cent of the total amount of home loans and at least 20 per cent of the total amount of loans. Home construction loan must not exceed 25 per cent of the total amount of loans.

The bank wishes to maximize its revenue from loan interest, subject to the above restrictions. The bank can lend a maximum of Rs 75 crore. You are expected to suggest to the management of the bank for maximizing revenue keeping in view operational restrictions.

Case 4.2: Accurate Transformers

Accurate Transformers produces electric transformers. The company has orders for transformer for the next six months. The cost of manufacturing a transformer is expected to somewhat vary over the next few months due to expected changes in materials costs and in labor rates. The company can produce up to 50 units per month in regular time and up to an additional 20 units per month during overtime. The costs for both regular and overtime production are shown in the table below:

	Month					
	Jan.	*Feb.*	*Mar.*	*April*	*May*	*June*
Orders (units)	58	36	34	69	72	43
Cost per unit at regular time (in '000 Rs)	180	170	170	185	190	190
Cost per unit at overtime (in '000 Rs)	200	190	190	210	220	220

The cost of carrying an unsold transformer in stock is Rs 25,000 per month. As on January 1 the company has 15 transformers in stock on and it wishes to have no less than 5 in stock on June 30. Suggest to the management of the company to determine the optimal production schedule.

Case 4.3: Nationwide Air lines*

Nationwide Airlines, faced with a sharply escalating cost of jet fuel, is interested in optimizing its purchase of jet fuel at its various locations around the country. Typically, there is some choice concerning the amount of fuel that can be placed on board any aircraft for any flight segment, as long as minimum and maximum limits are not violated. The flight schedule is considered as a chain of flight segments, or legs, that each aircraft follows. The schedule ultimately returns the aircraft to its starting point, resulting in a 'rotation'. Consider the following rotation: Delhi – Hyderabad – Cochin – Chennai – Delhi

Fuel Requirements and Limits (1,000 gallons unless specified otherwise)

City	*Flight Sequence*	*Minimum Fuel Required*	*Maximum Fuel Allowed*	*Regular Fuel Consumption if Minimum Fuel Boarded*	*Additional Fuel Burned per Gallon of Tankered Fuel (i.e. fuel above minimum – in gallons)*	*Price per Gallons (Rs)*
1.	Delhi to Hyderabad	23	33	12.1	0.040	8,200
2.	Hyderabad to Cochin	8	19	2.0	0.005	7,500
3.	Cochin to Chennai	19	33	9.5	0.025	7,700
4.	Chennai to Delhi	25	33	13.0	0.045	8,900

The fuel for any one of these flight segments may be bought at its departure city, or it may be purchased at a previous city in the sequence and 'tankered' for the flight. Of course, it takes fuel to carry fuel, and thus an economic trade-off between purchasing fuel at the lowest-cost location and tankering it all around the country must be made.

In the table above the column 'Regular Fuel Consumption' takes into account fuel consumption if the minimum amount of fuel is on board, and the column 'Additional Fuel Burned' indicates the additional fuel burned in each flight segment per gallon of 'tankered' fuel carried; tankered fuel refers to fuel above the minimum amount.

The fuel originally carried into Delhi should equal fuel carried into Delhi on the next rotation, in order for the system to be in equilibrium.

If l_1= Leftover fuel inventory coming into city i (1,000 gallons) and x_i = Amount of fuel purchased at city i (1,000 gallons) then, $(l_i + x_i)$ is the amount of fuel on board the aircraft when it departs city i.

Suggest the optimal fuel quantity to be purchased by the Airlines.

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* This case is based on 'D. Wayne Darnell and Carln Loflin, *National Airlines Fuel Management and Allocation Model*, Interfaces' Feburary, 1977.